

**How Does the Integration of Personalized Recommendations
and Experiential Learning Within Learning Technologies
Enhance African American Males' Understanding of Career
Pathways and Increase Engagement in Career-Focused
Learning?**

BY

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1 Introduction

1.1 Overview

Bridging the Gap in Career Exposure: A Technological Approach to Personalized and Adaptive Learning

The landscape of education is rapidly evolving, with a growing emphasis on aligning academic learning with future career success (Michael Healy et al., 2020; R. Honra et al., 2024; Zainul & Maskur, 2024). This shift has been driven by an increasing demand for career readiness, where educational systems are expected to prepare students not only academically but also for specific career pathways. In recent years, policymakers, educators, and industry leaders have called for a stronger connection between academic curricula and the skills required for success in the labor market. As traditional models of education struggle to keep pace with the rapid advancements in technology and the dynamic nature of job markets, career-focused learning has emerged as a potential solution. However, this shift towards career-focused learning raises questions about its effectiveness, particularly as it relates to students' preparedness for making informed career choices. Many students face challenges in navigating complex career options, often due to a lack of adequate guidance or exposure to diverse career paths early in their education. A notable gap in career preparedness (I George & Paul, 2024; Mazzotti et al., 2024) underscores the need for a more structured, career-focused approach that equips students with the necessary tools (Caratozzolo et al., 2024; Singha & Singha, 2023) and knowledge to navigate complex career decisions (Cureton & Aguinaldo, 2024; Slaten et al., 2024). For example, a survey of 582 students revealed that half of those who had chosen a career path expressed discomfort with their decision, struggling with indecisiveness and insufficient knowledge about occupations and decision-making processes, much like their undecided counterparts (Arbona et al., 2024). In contrast, research on career-focused curricula has demonstrated that such approaches deepen students' knowledge of career options (Jones, 2020; Zainul & Maskur, 2024) and have been effective in encouraging junior high school students to pursue IT careers (Genrich, 2020), as well as increasing primary and secondary students' perceptions of STEM careers (Chen et al., 2024). Addressing these challenges requires not only career-focused curricula but also an educational design that reflects the diversity of student experiences and adapts to their evolving needs. Tackling these challenges is where personalized learning approaches play a transformative role. Scholars have explored the potential of incorporating culturally responsive design elements to create more inclusive learning environments that reflect students' unique cultural backgrounds (A. Y. Cheng et al., 2024; M.-M. Cheng et al., 2021). Personalizing student engagement by tailoring learning experiences to individual needs (Nazempour & Darabi, 2023; Villegas-Ch et al., 2024) and integrating learning tools that adapt to both formal and informal learning contexts based on students' distinct characteristics (Contrino et al., 2024) have shown to increase student engagement. It may be interesting to explore how the combination of personalized learning integrating culturally responsive design and adaptive learning tools, like formal and informal learning, impacts not only a more engaging experience but an enriching and holistic career learning experience.

1.2 Motivation

Persistent concerns have arisen regarding the effectiveness of college and career readiness programs implemented by local and federal governments, particularly in their ability to prepare at-risk students, especially African Americans, for postsecondary education and career success (Raines & Talapatra, 2019). Despite educational initiatives like the Improving American School Act of 1994 (IASA) and the No Child Left Behind Act of 2001 (NCLB), there is a disconnect between these initiatives and students' college readiness (Peurach et al., 2019). The Educational Policy Improvement Center defines college and career readiness as the ability to “succeed in entry-level, credit-bearing college courses leading to baccalaureate or career pathway-oriented training programs without the need for remedial or development coursework” (Conley, 2012). Nevertheless, according to (Reed et al., 2024) only 22.6% of Black Americans earned a baccalaureate degree, 64% of Black Americans who earned a baccalaureate degree were Black females, and 50% of Black males without a high school degree were either unemployed or incarcerated.

African American boys face unique systemic barriers related to race, socio-economic status, and educational inequity (Lateef et al., 2024, 2024), that transcend various places, systems, and domains. While many career-focused tools promise adaptability and accessibility, they often fail to account for barriers that disproportionately affect marginalized communities. African American boys face challenges from a lack of access to resources, and from platforms that are not culturally responsive or reflective of their lived experiences (Ersoy, 2023; Lateef et al., 2024). This disconnect between educational technology design and the specific needs of African American males reinforces existing inequities, leaving them underprepared for postsecondary education and career opportunities (Lateef et al., 2024).

To dismantle these inequalities in the design of career-oriented technology, it is essential to include African American boys in the design process, ensuring their lived experiences inform the development of platforms meant to serve them. As (Egede et al., 2024) emphasize, “systems that are intended for a group should be developed by—or at the very least include—the group in the process.” Tools, such as technology that actively respond to the needs and realities of African American boys should ensure equitable access and help to disrupt systemic biases encoded into mainstream educational platforms (Benjamin, 2019; Costanza-Chock, 2020). I posit that co-design is essential for developing tools for African American boys. However, designing technology that may feel prescriptive, combined with engaging vulnerable populations historically subjected to bias research, necessitates precautionary measures before direct interaction (Harrington et al., 2019). Therefore, the initial phase should focus on creating a baseline model to test for potential systemic biases. By subsequently applying iterative user testing, these technologies can be refined with direct input from users, ensuring that the final outcome authentically reflects their experiences and needs.

Career-driven learning technologies must be intentionally designed with equity at the forefront, challenging the systems of power. Equity design requires going beyond surface-level inclusivity to create tools that are deeply aligned with the cultural, academic, and social realities of African American boys. Addressing these realities means confronting hidden biases embedded within educational technologies. As Ruha Benjamin argues in *Race After Technology*, algorithms often operate under the guise of objectivity, perpetuating systemic inequities through what she terms the New Jim Code (Benjamin, 2019; Dancy & Saucier, 2022). These technologies, which appear neutral, can, in fact, reinforce educational and career barriers for marginalized communities. Insights from the field of Design

suggest that career-focused tools should take an HCI approach that considers not only user interactions but contextual factors, such as geographical location and opportunities for greater representation in underexplored career paths. These are factors that perpetually limit marginalized students' progression and continuously reinforce systemic racism (Egede et al., 2024).

Ultimately, the goal is to transform learning technologies into spaces of empowerment where African American boys can fully realize their academic and professional potential. Early research that I conducted on high school students revealed the significant role of educational initiatives in helping students envision and take practical steps toward their career goals. This research emphasized the importance of experiential opportunities that allow students to explore and affirm their future career identities and that will directly inform the design of learning technologies.

1.3 Research Audience and Positioning

Learning technology serves as a key tool for addressing career readiness, as its scalability and adaptability make it ideal for expanding career exposure. However, technology alone cannot provide a holistic experience that supports and encourages agency and empowerment. The complexity of this exploration requires a multidisciplinary approach, drawing from design, education, and technology, building on existing scholarship while advancing these fields through the creation of new insights. To emphasize the magnitude of my work, the stakeholder map in Figure 1 illustrates the ecosystem that influences the components impacting this endeavor and identifies contributors whose influence may extend beyond the scope of my Ph.D. trajectory. The primary focus centers on four key contributors: African American males, teachers, administrators, and parents. These contributors play a central role in shaping and implementing the research outcomes, ensuring their relevance and impact within their respective domains.

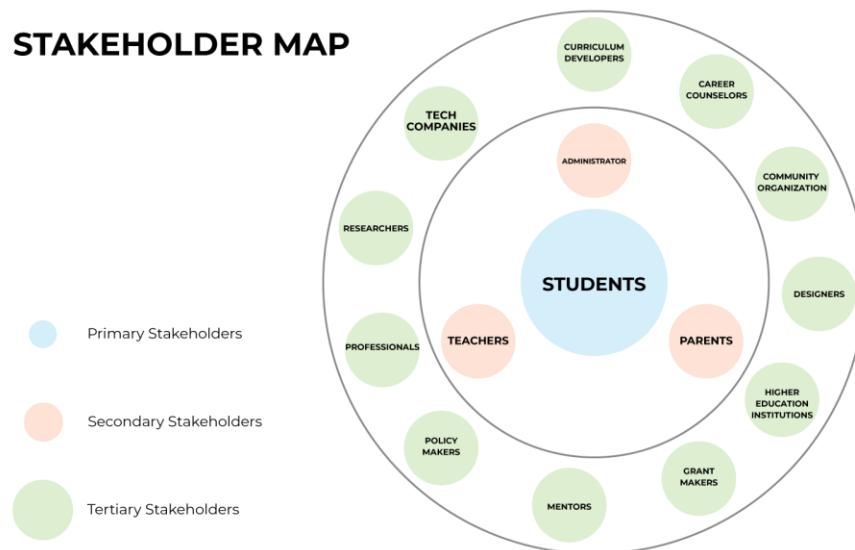


Figure 1 - Stakeholder Map

One question my research may raise but not fully address is whether technology could serve as a fifth key contributor. Although a diverse set of actors are positioned as the primary contributors, adaptive tools like artificial intelligence (AI) have the potential to act as knowledge producers. AI systems, for instance, can analyze student profiles, interact dynamically with students, and adapt based on feedback from teachers and parents, thus playing a significant role in shaping learning experiences. Its capacity to learn from these interactions may warrant its consideration as a contributor to the collective knowledge process, as described by (Jadallah et al., 2024), who state, “the idea of learning as a collective process in which a diverse set of actors interact and engage with each other to co-construct knowledge.”

African American Males in Marginalized Communities in Grades 6th through 8th

As shown above in my stakeholder’s map, African American males (AAM), grades 6th through 8th, are the key stakeholders. AAMs are positioned within communities facing systemic barriers that limit their academic and career opportunities (Thiem & Dasgupta, 2022; N. Thompson et al., 2023). To effectively address the barriers that hinder AAM from obtaining sufficient information and preparation for making informed academic and career decisions, it is crucial that learning technologies adapt to their unique needs and provide genuine engagement with material that is both relevant and accessible (De Caro-Barek et al., 2024), and that provide deep conceptional learning (Sawyer, 2022).

Historically, AAMs have been excluded from the design process, which explains issues such as facial recognition technology misidentifying them (Bacchini & Lorusso, 2019; Benedict, 2022), and discriminatory search engine results against African Americans (Noble, 2020), both of which have contributed to mistrust among AAM. To rebuild trust within this community, technologists should borrow from the design principles and ensure that educational technology is designed with context and scaffolding. The proposed approach posits that recommender systems with a direct impact on students allow them to shape the tools by incorporating student models that reflect their aspirations, address their challenges, and understand the impact of technology (De Caro-Barek et al., 2024). As shown in Figure 2, the model explores how to incorporate students' unique imprint by factoring in personalized information, including hobbies, interests, lived experiences, and feedback. The student model will be covered in more detail in later chapters, but it must be noted that the student model is designed to evolve, adapting to reflect student’s growth and changing career goals.

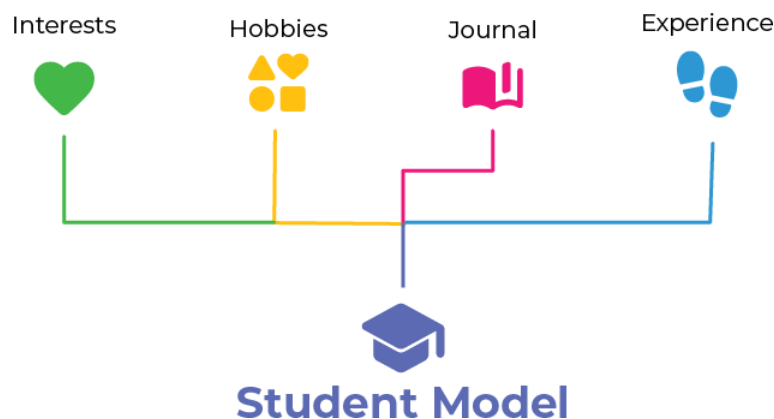


Figure 2- Student Model

Parents of African American Boys in Marginalized Communities

Parents are essential contributors and are positioned as key contributors on the above stakeholder's map, offering valuable insights into their children's aspirations and challenges (Koçak et al., 2021; N. Thompson et al., 2023). Parents' feedback on their children's strengths, weaknesses, and potential career trajectories is instrumental in assessing parental perspectives on their child's future, and research shows that it has a positive effect on students (Zahra & Malik, 2024). Parental feedback is an important data asset that is ingested into the student model, as shown in Figure 3. This additional layer of data has the potential to normalize data points in the model that may be skewed, such as academic information affected by biases in grading systems. Details on how parental feedback functions within the student model will be covered in later in my dissertation.

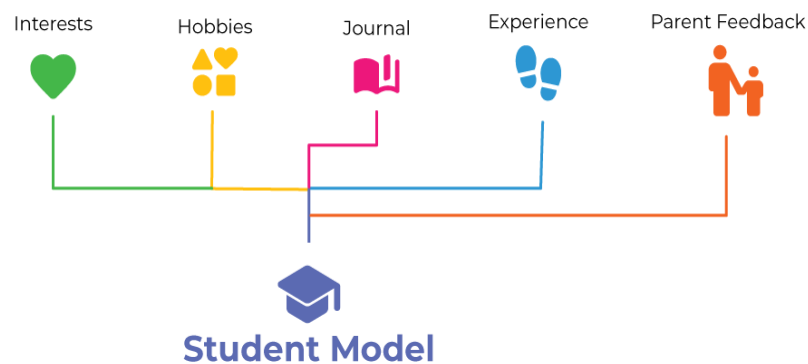


Figure 3- Student Model (Parent Feedback)

Parents are active agents in the collaborative process. This collaboration is multi-dimensional, enabling parents to understand their child's career paths and development, and students can benefit from their parents' insights. Technology also acts as an adaptive agent, learning from both parent and student interactions, while parents and students, in turn, are influenced by technologies that integrate adaptability tools to support evolving learning needs.

While collaboration between parents and students can enhance the learning process, it may simultaneously generate tensions. Drawing on the insights from interdisciplinary evaluations of AI systems (Z. Chen & Schmidt, 2024; Zicari et al., 2021), it becomes evident that ensuring transparency and trust in how information is shared between parents and students is crucial. This collaboration may raise significant ethical and privacy concerns that must be carefully addressed when designing solutions that engage multiple stakeholders in authoritative relationships. Further details on ethical and privacy concerns will be provided in Section 3.3, with an emphasis on maintaining student agency by allowing parental access to certain information while safeguarding student privacy.

Teachers and Administrators

Teachers and Administrators are key contributors, as shown in the stakeholder map above, and this co-collaboration between teachers and students plays a crucial role in this process. Teachers are the main point of contact for students in their learning ecosystem outside of their parents. Due to the amount of time teachers interact with students, they are able to provide qualitative insights that are sometimes not reflected in student grades. These insights allow teachers to design more individualized learning experiences tailored to the unique needs and career trajectories of each student (De Caro-Barek et al., 2024). The dynamic exchange of information between students and educators creates a responsive and adaptive educational environment that supports both academic growth and career preparation. Both teachers and administrators play a key role in supporting the implementation of learning technologies. The outcome of this collaboration can be integrated into supportive learning environments that reflect the cultural and individual needs of marginalized students (De Caro-Barek et al., 2024). Through this process, educators can provide valuable feedback on the technology's effectiveness and alignment with student needs.

Teachers are also provided with a plethora of data-driven tools to assist them in creating effective learning experiences for students. That data is leveraged and indicated as ‘Academic’ in the student model. The academic data assets will be discussed more in the method section.

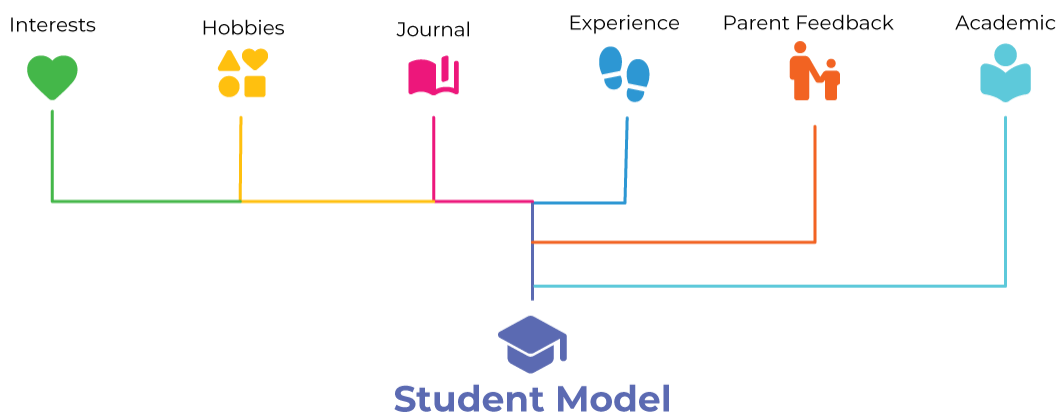


Figure 4- Student Model (Parent-Teacher)

1.4 Research Questions

My research goal is situated at the intersection of education, technology, and design. It explores how combining personalization, adaptability, and collaboration can provide students with opportunities to engage with relevant content and experiences. The anticipated outcome is to leverage technology to empower students by enhancing career awareness while simplifying the complexity of career pathways through career exploration. Using a prototype-based approach, my study proposes that a career-focused approach—utilizing interest-based analysis and adaptive learning to create formal and informal learning experiences—will broaden students’ understanding of career options and empower students, particularly those from marginalized communities, to make informed career choices, ultimately preparing them for successful futures in a rapidly changing workforce.

Situating my research: The interplay of personalization, adaptability, and collaboration across, education, technology, and design

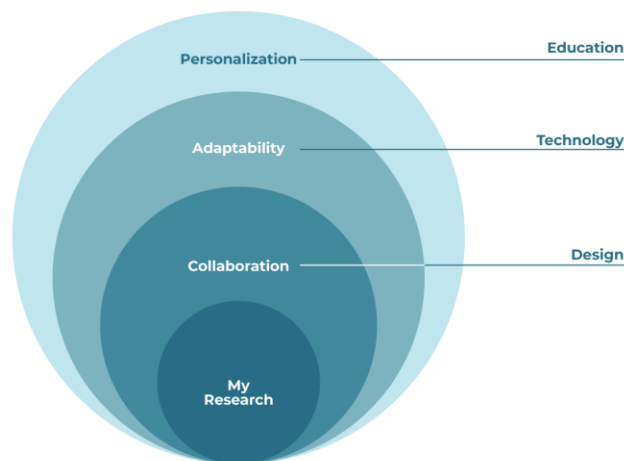


Figure 5 - Situating My Research

My proposed main and sub-questions are as follows:

Main Question: How does the integration of personalized recommendations and experiential learning within learning technologies enhance African American males' understanding of career pathways and increase engagement in career-focused learning?

Sub-Questions:

RQ1: How do interest-based personalized recommendations within learning technologies influence career awareness among African American male students?

RQ2: How do formal and informal experiential learning activities within career-focused learning technologies impact African American male students' career exploration?

RQ3: How does incorporating geographical data from the ACS aid in recommending careers aligned with students' interests, particularly those where African American males are underrepresented?

2 Literature Review

2.1 The Career-Focused Approach

The goal of our educational system is to equip students to realize their full career potential. To meet the needs of American students, national education reform initiatives were established to offer a structured framework for their development. In 2010, the Obama administration set two educational goals: to raise the college graduation rates and to close the academic gap so high school students are ready to succeed in college and careers (Cristol et al., 2015). In 2015, The Every Student Succeeds Act (ESSA) focused on college and career readiness (CCR) frameworks (Malin et al., 2017). These initiatives were designed to elevate student expectations and promote the use of standards and assessments to evaluate students' competency levels at the time of graduation.

The concept of being college and career-ready is centered on ensuring that students are adequately prepared, starting from high school graduation and continuing into the workforce. Career Counselors play a pivotal role in connecting students' education experiences and exposure to their future (S. Chen et al., 2021; Curry et al., 2013). There is a critical need for occupational guidance, as discussed by Wilmot and Ones (2021), who suggest that career guidance helps address challenges related to self-confidence and the need for external validation in decision-making scenarios. Emphasizing the development of informed career decision-making as a crucial 21st-century skill is essential for promoting students' career-related performance in today's occupational market. The effectiveness of career interventions not only benefits individuals but also offers valuable insights for policymakers and guidance practitioners, highlighting the indispensable role of career guidance in today's rapidly changing world, as suggested by (Chiesa et al., 2016). Research shows that career counseling enhances students' academic motivation (Nye et al., 2021), provides individual assessment and career-education matching (Firkola, 2021), and increases their interest and readiness for the future (Lau et al., 2020). In the United States, the U.S. Department of Education recommends a student-to-counselor ratio of 430:1 (Dutta et al., 2022), well above the American School Counselor recommendation of 250:1 (American School Counselor Association, 2015). This large disparity often exacerbates the challenges faced by underserved students, who may require more intensive support and guidance to navigate their academic and career pathways effectively. In many schools, especially those serving low-income and minority communities, the actual ratios, 1000:1, far exceed the recommended level (Zahrebniuk, 2023), despite literature (Brown & Knight, 2023; Bryan et al., 2022; Hilts et al., 2023) showing a positive impact when those ratios are low, limiting the availability and accessibility of personalized counseling services. This situation can hinder these students' ability to receive the necessary guidance for college preparation, career exploration, and social-emotional support, thereby widening the gap in educational equity and opportunity. The lack of educational resources provided to these students may explain the prevalence of career indecision among students, in addition to other drivers, such as the misalignment between abilities and interests and a lack of self-knowledge, as identified by (Dozier et al., 2022).

The integration of Artificial Intelligence (AI) into career guidance education represents a transformative shift in how students across the globe navigate their career paths. The article from the *European Journal of Artificial Intelligence and Machine Learning* by (Yao Cheng & Yu Liang, 2023) provides a comprehensive overview of this evolution, highlighting the historical development, application areas, and the multifaceted impact of AI on career exploration for

children and adolescents. Cheng and Liang trace the origins of AI in career guidance back to the 1960s, marking a departure from traditional pen-and-paper assessments towards more interactive and personalized tools. Early AI-assisted functions aimed at aligning adolescents' interests and skills with suitable career choices, albeit with limitations in interactivity and adaptability to changing job markets. This historical perspective underscores the gradual evolution of AI tools from rudimentary algorithms to sophisticated systems capable of offering immersive career exploration experiences. This progression has culminated in AI's current capability to offer dynamic and personalized career guidance solutions, far surpassing the early limitations and setting a new standard for how career advice is administered.

This theoretical potential of AI is grounded in empirical evidence, as demonstrated by the research conducted by (Arafath et al., 2018; Comendador et al., 2016; Neelamani et al., 2023), which collectively showcases the practical efficacy of machine learning (Shelehov et al., 2023) in refining the process of matching students with optimal career paths. Research by (Arafath et al., 2018) at Daffodil International University exemplifies leveraging AI for personalized career guidance, aligning with the rapidly evolving job market. The study aimed to predict the future career paths of current computer science students by analyzing data from successful alums. The analysis focuses on key parameters such as professional skills, interpersonal skills, and academic records to make accurate predictions. (Arafath et al., 2018) used multiple classification algorithms to predict an estimated career for a student. Classification algorithms, defined by (Marinai, 2008), are like skilled navigators charting the course through uncharted waters, using past patterns to guide future journeys. (Arafath et al., 2018) leverages this technique to predict and track students to understand academia-industry collaboration better. (Neelamani et al., 2023) study focused on the application of machine learning models for predicting suitable job profiles for students based on their academic and technical performance and showed that random forest classification demonstrates a high accuracy rate in matching students with job profiles that align with their academic and technical skills. This study addresses the growing need for accurate and predictive tools in campus placement through the application of machine learning (ML) models.

The exploration of machine learning and natural language processing (NLP) in career guidance, as discussed by (Yao Cheng & Yu Liang, 2023), offers personalized advice by analyzing vast amounts of data. This aspect is particularly relevant to research exploring how AI can tailor career guidance to individual profiles, thereby democratizing access to career opportunities. Creating a level playing field where marginalized students have an equal chance to explore career opportunities without facing barriers and providing intuitive career information that allows students to make informed decisions about their career paths. NLP, a branch of artificial intelligence, focuses on the interaction between computers and humans through natural language by understanding, interpreting, and generating human languages that are valuable and meaningful. NLP can perform several tasks, such as categorizing text into predefined categories (Mittal et al., 2020), translating text from one language to another (Wang et al., 2023), identifying and classifying key elements in text into predefined categories (Tan et al., 2023), building systems that can answer questions (Panditharathna & Rajapaksha, 2023), and generating a concise and coherent summary of large documents (Mishra et al., 2023). Language Models (LM) at the foundation solve these various NLP tasks. (Bender & Koller, 2020) define language models as systems that are trained in string prediction tasks that predict tokens (string, word, or character) given either its preceding or surrounding context.

2.2 John Holland's Theory and Its Application

AI's ability to operate at scale offers a promising solution to the lack of resources in schools in marginalized areas. However, it is crucial to integrate theories to justify the methods used to match students to careers (Koys, 2017). According to (Hussain et al., 2021; Nye et al., 2021; Su, 2020), personality and interests are key determinants of academic and career success. Many scholars have explored this theme: (Darley & Harenah, 1955) proposed that interests in various work activities reflect what people find meaningful or fulfilling; (Layton, 1958) believed interests represent a specific component of a person's overall motivation; and (Hansen, 1994) stated that personality and interest variables are related and measurable. However, it was John Holland who proposed a sixfold taxonomy of individuals and occupations (J. L. Holland, 1966). Holland's theory provided a framework for organizing occupational interest data and aligning individuals' personality types with compatible work environments, thus enhancing career satisfaction and productivity

My research builds on John Holland's personality trait theory, which dates back to 1966. Between 1966 and 1997, Holland developed, defined, and provided interests and competencies to six primary personality traits that people are classified as (Realistic, Investigative, Artistic, Social, Enterprising, and Conventional), which is described in the figure below (J. Holland, 1997; J. L. Holland, 1966; J. L. Holland & Gottfredson, 1976; SUPER, 1985). Holland believes that vocational interests are "an expression of personality" (Hogan & Blake, 1999).

My position is not to suggest that a student fits into only one personality type. Instead, it uses Holland's theory to help students focus on a smaller set of careers that match their traits. This approach makes it easier for students to explore their options without feeling overwhelmed by the countless career paths out there, which can often lead to confusion and a lack of preparation.



Figure 6- RIASEC Personality Description

Holland's theory also affirms that environments can be classified into the same six categories, which are described in the figure below. Each personality type aligns with a specific environment shaped by the people who primarily work in it. For instance, investigative environments are led by Investigative individuals and demonstrate the development of their unique attitudes, interests, values, and skills.

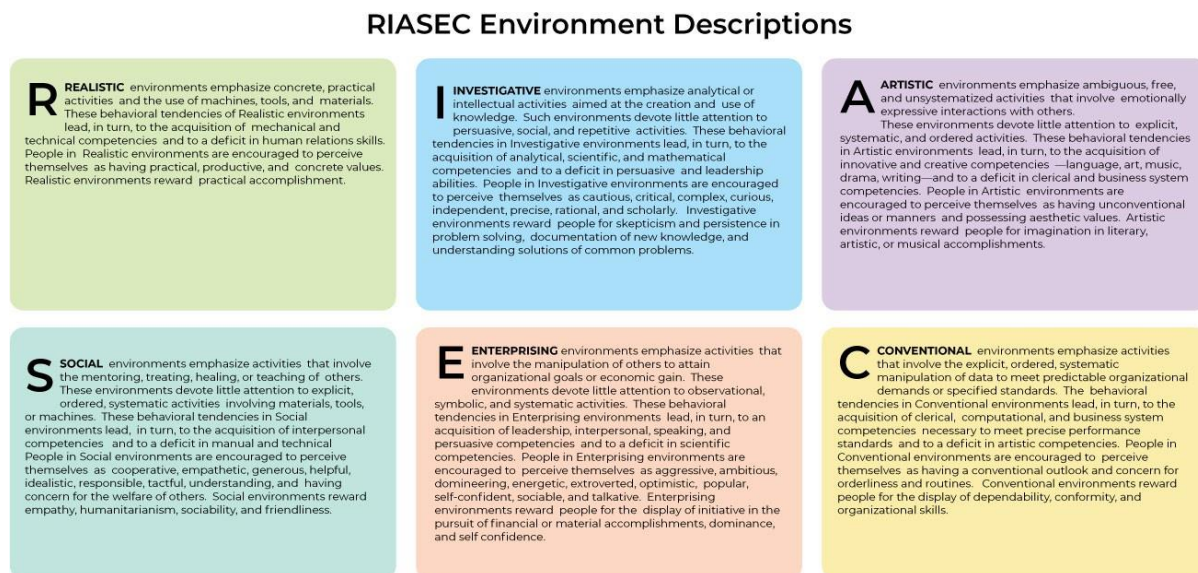


Figure 7- RIASEC Environment Description

Holland's theory posits that individuals will thrive in environments that align with their personality type, a concept he terms the person-environment fit theory (J. Holland, 1997; Koys, 2017). His theory has been widely influential, with over 19,015 citations, as per Google Scholar. According to (Putka et al., n.d.) the RIASEC model is the foundation for interest information due to its extensive use in applied settings and research.

Instead of focusing on individual traits and behaviors, which is common in psychology, Holland considers work environments and career interests (J. L. Holland, 1966). Existing research demonstrates that Holland's RIASEC theory effectively aligns careers with personality traits and interests (Day & Rounds, 1998; Tracey & Rounds, 1993). Holland created a simple way to show how well different personality types match with specific work environments. A hexagon was used to arrange six types of personalities and environments in this order: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional, as shown in the figure below. The closer types are on the hexagon, the more similar they are. For example, Realistic and Investigative types are right next to each other, so they are more alike, while Realistic and Social types, which are farther apart, are less similar. The arrangement of these RIASEC types in a hexagonal order has been supported by researchers, dating back to foundational studies in the field, such as those conducted in 1993 (Tracey & Rounds, 1993) and 1998 (Day & Rounds, 1998), underscoring the historical significance and long-standing validation of this theory.



Figure 8 - RIASEC Model

The model also shows how well people fit into environments. A perfect match happens when a person's type is the same as their environment—like an Investigative person in an Investigative environment. If an Investigative person works in an Artistic or Realistic environment, Holland's theory suggests that it is a good fit but not perfect. However, an Investigative person in an Enterprising environment is not a good fit at all. This hexagon model visually explains how people and environments can best work together, which is central to Holland's theory.

Holland's theory identifies a variety of ways of assessing a person's personality traits using qualitative and quantitative methods. One of the qualitative methods is observing whether a person prefers or is actually working in a job or pursuing a career that aligns with a particular personality type. (J. Holland, 1997). In collaboration with Holland, there have been efforts made to identify occupations with personality types, such as The Dictionary of Holland Occupational Codes (DHOC) (Gottfredson, 1996), which has been adopted by the US Department of Labor and will be discussed in more detail in the next section. (Rosen et al., 1994) classified over 900 college majors according to how they align with the interests, skills, and abilities of each personality type.

Similar to the qualitative method used above, Holland's colleagues used a multitude of approaches that offer approaches to assess how well a person's traits align with specific career-related factors. These methods measure personality traits, activities, occupational preferences, competencies, and self-assessment scales from frameworks like Holland's six personality types (J. L. Holland, 1985; J. L. Holland et al., 1997) and the six theme scores from (Campbell & Hansen, 1981). Holland advocated for using these scales to establish a person's interests and traits as well as their fit with occupations and academic paths, which he considered the ideal approach (J. L. Holland et al., 1997).

Holland's theory has been widely applied across various fields. For example, (Pardi, 2024; Salako & Adediran, 2024) used it to explore career readiness among deaf children, aiming to create a more inclusive environment for their professional growth. (Fernández-Nistal et al., 2024) applied it to study the alignment between personality types and work environments among 274 students in Mexico, focusing on the Big Five personality traits and career confidence (vocational self-efficacy). (Taqyah et al., 2024) used the model with 1,609 high school students to validate the RIASEC model and found no differences in validity by gender or academic major. Research in Indonesia showed that secondary students' vocational personality profiles varied based on their location and school reputation, highlighting the need for

counselors to consider demographic factors when providing career planning services (Faculty of Teacher Training and Education, University of Lampung, Indonesia & Dahlan, 2024). (Dinampo et al., 2024) examined the correlation between personality types and curriculum choices among 12th graders in Cavite, ultimately recommending that students take reliable career assessments, like RIASEC, to help them identify careers matching their personality traits, especially those aligned with Social, Conventional, or Realistic pathways. According to (K. R. Thompson & Koys, 2010), this model was used to map out a plan for curriculum development and assessment, (Freeman & Hirsch, 2008) used it to map it to college majors, and (Robinson et al., 2000) used it to create an informative education to career database.

2.3 The Role of Holland's Theory in U.S. Career Frameworks

The RIASEC typology has proven to be highly accurate, leading to its integration into the U.S. Department of Labor's unemployment model. The Occupational Information Network (O*NET), developed by the U.S. Department of Labor, serves as a comprehensive resource detailing information on over 900 occupations within the U.S. economy (Koys, 2017). Holland's theory, which posits that individuals are more satisfied with their occupations when they align with their interests, forms the basis for much of O*NET's structure. The system is grounded in the O*NET Content Model, which serves as its conceptual framework (Mumford & Peterson, 1999). This model organizes critical occupational data into a structured system that is both theoretically sound and empirically validated.

Derived from extensive research in job and organizational analysis, the content model captures the core of occupations through job-oriented descriptors and represents worker traits through worker-oriented descriptors. It applies information across various jobs, sectors, or industries via cross-occupational descriptors, as well as within specific occupations through occupation-specific descriptors. As shown in the figure below, these descriptors are grouped into six major domains, enabling users to identify essential attributes and characteristics of both workers and occupations. The information provided in O*NET includes key details such as tasks, knowledge and skill requirements, wages, and interests for each occupation. Additionally, each occupation in the database is rated based on the RIASEC types, facilitating systematic person-career matching (Converse et al., 2004).

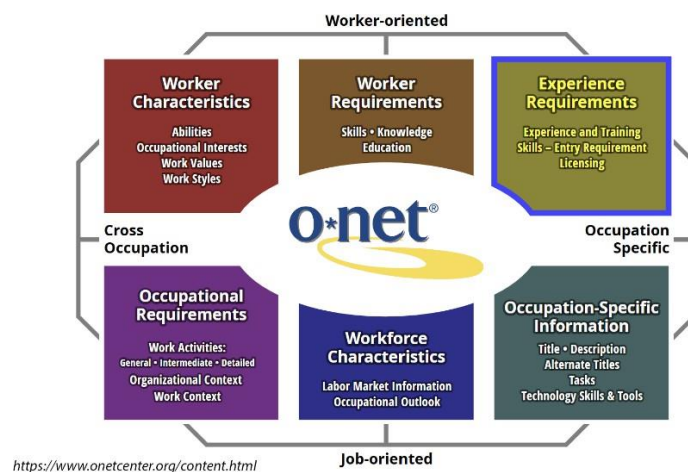


Figure 9 - O*Net Content Model

Despite the effectiveness of Holland's theory in aligning careers with personality traits, it becomes essential to recognize that cultural factors significantly influence how students perceive and engage with career opportunities. Theoretical frameworks like RIASEC, while robust, do not inherently account for the unique cultural backgrounds that shape students' values, interests, and aspirations. As AI and machine learning continue to advance the personalization of career guidance, integrating culturally responsive design becomes increasingly important to create career pathways that resonate with students from diverse backgrounds, particularly those from underrepresented communities.

2.4 Exploring Adaptive Learning Systems

Adaptive learning systems tailor educational experiences to fit the individual needs of learners. These systems adjust to diverse preferences, performance levels, and goals. However, despite advancements, gaps remain in providing interest-based and culturally relevant personalization and adapting learning across formal and informal environments. The following narrative synthesizes findings from four key studies on adaptive learning systems and examines how these gaps can be addressed through the use of context and scaffolding specifically for African American boys.

Personalization: Interest-Based and Culturally Relevant Content

In adaptive learning, several systems tailor content based on academic performance, but they often miss opportunities to align learning with students' personal interests and cultural backgrounds. In the study by Contrino et al. (2024), the adaptive learning tool functioned like a one-size-fits-all suit. It adjusted content delivery based on student performance metrics, such as grades and test scores, but lacked cultural context by failing to account for individual interests or cultural backgrounds, especially for students from marginalized communities.

For instance, consider a student named Jamal, a 7th grader with a passion for healthcare. While the system helps Jamal improve his grades in science, it does not connect his learning to the stories of Black medical pioneers or show him how his love for helping others could turn into a fulfilling career. Jamal is unable to contextualize the material in relation to his future career self, leading to a disconnection from the content.

This is a lost opportunity to scaffold the student by guiding Jamal to make a deeper connection with the material, and it is common in educational settings (as illustrated in my preliminary findings in section 4.1.4). (Demartini et al., 2024) expanded personalization to include real-time feedback on academic performance; however, the main focus was on academic metrics, and they lacked interest-driven or culturally relevant learning. Mejeh and Rehm (2024) presented an adaptive system that functioned like a GPS, guiding students through academic progress but missing personalized paths based on student interests or cultural relevance. In contrast, (Curry et al., 2013) demonstrated how O*NET could serve as a compass, aligning students' career interests with their learning experiences.

For Jamal, O*NET would guide him toward healthcare careers that feature African American pioneers in medicine, helping him see himself in the profession. Cultural framing makes learning more relevant and meaningful to his personal and cultural identity.

The first three studies treated personalization like a machine on autopilot, missing the opportunity to engage students through their interests and cultural context. My work picks up this thread, using O*NET and Holland's RIASEC theory to create tailored, culturally relevant learning experiences.

For example, Jamal would be provided with learning resources about Black doctors and healthcare leaders, connecting his passion for medicine with his studies, which would reignite his interest in learning.

Adaptability: Formal and Informal Learning

While personalization focuses on tailoring the content of what students learn, adaptability deals with how and where learning occurs. (Contrino et al., 2024) focused on adapting content within formal university settings, similar to a rigid syllabus that does not consider the fluid nature of informal learning environments such as internships, after-school clubs, or community activities.

Take Elena, a middle school student who excels in formal classroom settings but thrives when learning through hands-on experiences. She loves engineering and spends her weekends tinkering with DIY projects at home. However, her adaptive learning system does not account for this informal learning, and her love for engineering is not nurtured in class. The system only adapts to her academic needs, missing the opportunity to connect her formal education to her real-world interests.

(Demartini et al., 2024) made learning adaptable in the classroom, but much like a plant constrained by a pot, the system did not allow learning to grow beyond the boundaries of the classroom walls. (Mejeh & Rehm, 2024) also kept adaptability confined to formal settings, focusing solely on the academic environment without accounting for the rich learning opportunities that exist outside.

On the other hand, (Curry et al., 2013) showed that O*NET can serve as a bridge, seamlessly connecting formal academic tasks with informal experiences such as community involvement and internships.

In Elena's case, there was an opportunity to scaffold her from her classroom lessons on engineering principles to real-world projects like shadowing a local engineer or participating in an after-school robotics club. This integration would make her learning more practical and meaningful.

While the earlier studies-built walls that confined learning to formal environments, Curry et al. (2013) demonstrated how O*NET can blend formal and informal learning. Building off this idea, O*NET can be used to cultivate a learning garden where students like Elena can grow through both classroom learning and real-world experiences. By connecting her interest in engineering with opportunities for informal learning, Elena would be better prepared for a future career while remaining engaged and motivated in school.

My research addresses gaps in personalization, adaptability, and collaboration by integrating O*NET and Holland's Theory. This framework creates a learning environment where students like Jamal and Elena are not only academically successful but also culturally engaged, career-ready, and prepared for real-world challenges. By ensuring that students receive personalized and adaptable learning experiences, the goal is to provide students with a deeper understanding of career pathways. This review underscores the need for inclusive approaches to learning technologies that empower

students to achieve their full potential by providing deeper conceptual understanding while emphasizing personalized and culturally relevant resources.

2.5 Gap in Knowledge

My work addresses a critical gap in how career exploration tools are tailored for African American boys by investigating how incorporating O*NET data and Holland's RIASEC theory can bridge these gaps by offering personalized, culturally relevant career recommendations and adaptive learning experiences that extend beyond the classroom. While many platforms exist to help students discover careers based on their skills and interests, these tools often fail to combine and incorporate cultural context and interactive and adaptive scaffolding. This lack of representation can hinder engagement and limit aspirations, as students may not see themselves reflected in various career paths or have access to tailored resources. My work aims to fill this gap by incorporating culturally relevant elements and personalized experiences that empower these students.

Although adaptive learning systems have made significant strides in personalizing content based on academic performance, there is still a notable gap in integrating interest-based and culturally relevant personalization, adaptability across both formal and informal learning contexts, and real-world collaborative experiences. Yew Kee Wong points out that a gap exists in aligning “education for employment, AI in education, and AI in employment” (Kee Wong, 2021). My research addresses “education for employment” by focusing on the connection between education and career preparation. Current adaptive learning tools emphasize academic success within the classroom while overlooking career readiness and real-world skills in informal settings. The existing body of research, despite being rich in AI-driven educational tools, does not fully integrate career exploration and guidance into the educational experience, creating a gap between what students learn and how they can apply that knowledge in real-world career scenarios.

Personalization in these tools typically focuses on academic performance without accounting for the students' unique cultural backgrounds and personal interests. This oversight is especially important for marginalized students whose career ambitions are often influenced by their cultural context and lived experiences. Without culturally relevant content and guidance, students may struggle to see how their education connects to future employment opportunities.

3 Research Methodology

3.1 Methods and Framework

As described in Section 1.4, the goal is to understand how career awareness (Anssi Salonen et al., 2018) and streamlining the complexities of career exploration (Arbona et al., 2024; Yao Cheng & Yu Liang, 2023) can empower students (M.-M. Cheng et al., 2021) by guiding them through career trajectories while adapting to their unique needs. Situated in the education, technology, and design disciplines, my work borrows from these interdisciplinary to create a personalized experience that evolves alongside the student's career growth.

Prototype-Based Research Method

My research follows a prototype-based research approach, drawing on the purposes defined by (Camburn et al., 2017), refinement, communication, active learning, and exploration. A career-focused approach is employed, incorporating previously discussed methods such as Holland Code, O*Net Content model, and student models. The personalized learning framework tailors content to each student's interests, academic background, and career aspirations, creating career recommendations. The experiential learning methodology complements this by providing students with opportunities to engage in hands-on activities that deepen their understanding of various career paths.

Design students were used as a proxy to develop a UX experience based on design research to create a design suitable for prototyping with my target audience. However, co-design serves as the bridge, giving students agency and empowerment to contribute to the prototype's design. Although these steps are not followed to the letter outlined in (Ege et al., 2024), they follow the refinement process. Through collaboration, students will work with the prototype, enabling their voices to be heard and integrated in a way that reflects their aspirations, cultural contexts, and personal challenges. Co-design is important as it provides identity awareness, enabling students to see themselves and their unique experiences represented in their educational journey (Egede et al., 2024). Section 5.2 discusses the prototype schedule in more detail.

The communication purpose, as described by both (Camburn et al., 2017; Ege et al., 2024) was realized through a summer project collaboration with design students, which is discussed later in Section 5.1. African American students are viewed as experts, and prototyping with them is designed to create a co-sharing of information. Students will have the opportunity to communicate their interests and hobbies, allowing me to develop the interest dataset artifact further. The active learning and exploration phases are expected to provide insights into whether students are engaging with the methods in this prototype. Validating students' learning outcomes will be validated later using pre- and post-assessments, observations, and interviews.

The following methods are proposed for the development of a working prototype, which will later be prototyped and tested with my targeted audience.

Career-Focused Model

My study builds off the O*Net Model discussed in Section 2.3, which aligns a person's RIASEC profile to an Occupational RIASEC profile to generate an Interest-Occupational Match. From a high-level perspective, this method entails two main processes, as illustrated in my figure below: Interest Profile (IP) and Occupational Interest Profile (OIP).

Interest Profile (IP)

The IP is based on a self-reported survey; O*Net has an interest profile survey that consists of 30 questions per RIASEC, totaling 180 questions; a short form that consists of 10 questions per RIASEC, which totals 60 questions, and a mini-interest profile form that consists of 5 questions per RIASEC totaling 30 questions (Gregory & Lewis, 2016). Research has shown that the correlation coefficient aligns for either survey, where the mini is more commonly used in time-sensitive scenarios (Gregory & Lewis, 2016). Research shows that long surveys are correlated with low participation, so the mini-interest profile survey is adopted, which, unlike the Department of Labor, which focuses on unemployed individuals, focuses on African American boys in grades 6th through 8th.

The Occupational Interest Profile (OIP)

The Occupational Interest Profile (OIP) uses a combination of three main methods to classify careers into Holland's RIASEC categories, which is shown in the figure below. A summarization of these methods as defined by (Rounds et al., 1999) is as follows: First, the judgment method enlists trained judges to manually classify jobs into RIASEC categories, refining or filling in gaps in the profile where other methods may fall short. Second, the empirical method uses statistical analysis to predict a job's RIASEC alignment based on detailed job descriptions, broadening classification beyond available worker data. The O*Net model primarily uses this method for developing OIP because it allows for a broad classification using statistical models. Lastly, the incumbent method collects data directly from workers in various fields to understand their RIASEC traits, using their profiles to shape occupational codes.

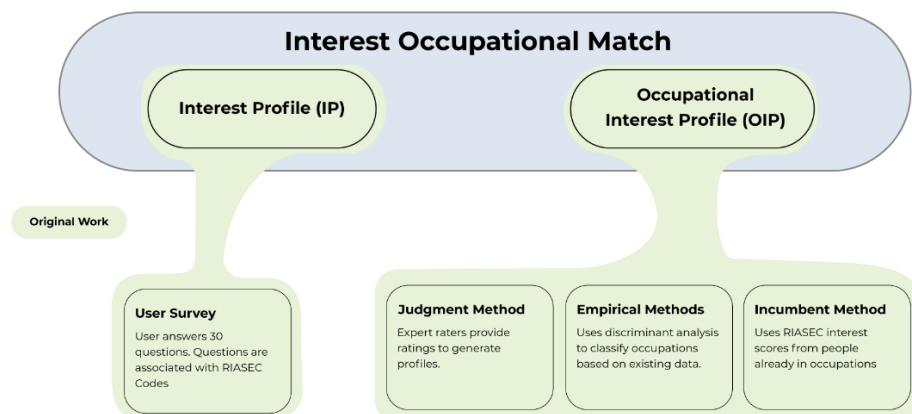


Figure 10 - Career Focused Model

Adaptations to the IP and OIP Models

My research employs the Interest-Occupational fit model, which matches a user's interest profile with an occupational profile as one persona in my prototype. However, as discussed further in the Personalized Learning section, additional fit criteria have been developed to align career recommendations with the needs of my targeted audience. Leveraging and extending the O*Net model and drawing from methods in the technology field to support career-focused learning, I undertook a custom data modeling and integration process by building the extensive O*Net Content Model into a structured SQL Server database. Using principles commonly applied in technology, I referenced the O*Net API's data model and replicated its structure and processes within SQL Server, following an approach similar to an ETL (Extract, Transform, Load) process (Khan et al., 2024). This technique involved extracting, transforming, and loading data to align with O*Net's career information model and testing my algorithm results to match O*Net's API outputs. This detailed work shown in Exhibit B serves as the foundation for my analysis, but more importantly, offers control over the data structure allowing my work to implement the personalized methods proposed in this research.

Once the data structure was established, mapping the IP to the OIP required replicating the Pearson correlation coefficients formula (below) into an algorithm to measure how closely the IP aligns with occupational profiles. Using the same matching criteria—"Very Strong Match ($\geq .729$)" and "Strong Match (.608–.729)" (Gregory & Lewis, 2016), I replicated the process to the O*Net career recommendation process, and these steps will be further discussed in my dissertation.

$$r_{XY} = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{N\sigma_X\sigma_Y}$$

Another factor that will be discussed in more detail in my dissertation is that although the O*Net database contains over 900 occupations, it does not include modern occupations, such as Social Media Influencers. A proposed approach is the use of Artificial Intelligence (AI) techniques, such as machine learning and natural language processing (NLP), to introduce newer careers to the career dataset without relying on O*Net methods, such as the Empirical Method. The proposed model, illustrated in the figure below, employs two methods: machine learning that uses historical job data to train a model that predicts RIASEC profiles for occupations, identifying careers where African Americans have been historically unrepresented; NLP, on the other hand, can analyze job descriptions and extract relevant RIASEC traits (skills, abilities, knowledge).

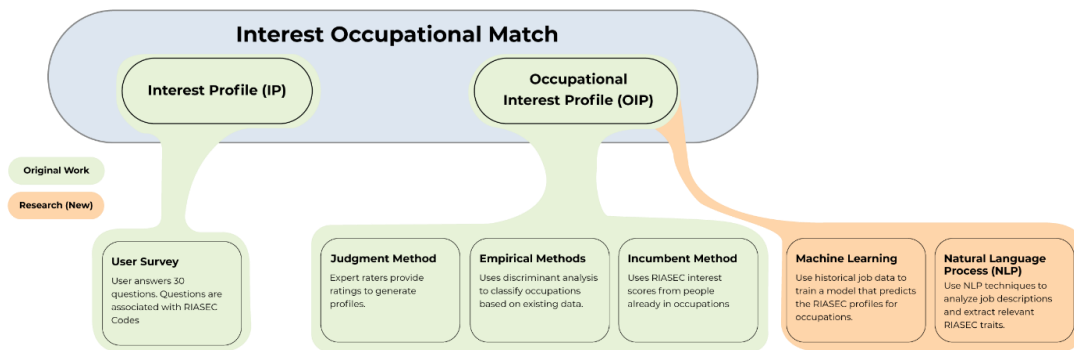


Figure 11 - Career-Focused Model - Research Approach

The process of matching careers based on interest-occupational fit occurs within a controlled environment. The following section discusses how additional factors, such as academic, cultural identity, and personalized learning pathways, are incorporated to create an impactful career exploration tool.

Personalized Learning

At the core of the personalized learning method is John Holland's Theory of Career Choice, which serves as the foundation for integrating these elements. As discussed in Section 2.2, Holland's theory believes the greater congruence between a person's interests and their career choice leads to increased success and satisfaction. My work builds off this model by adopting three approaches: scholastic fit, cultural opportunity fit, and personal fit, as shown in the figure below, to create a personal experience. Scholastic fit incorporates academically driven information, while cultural opportunity fit explores the method of applying high coefficients to careers in which African American males are underrepresented. Personal fit aligns more closely with the student's interests, hobbies, and journaling data points.

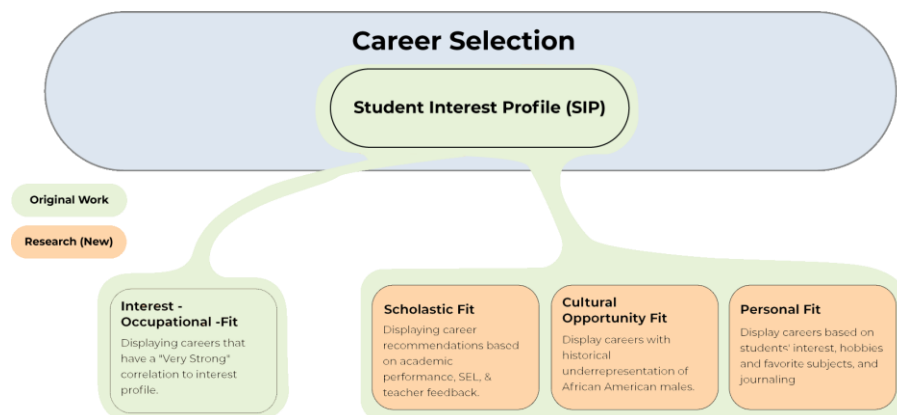


Figure 12- Personalized Experience - Career Selection

Scholastic Fit

The goal is to tap into some of the data-driven platforms that educators use to create a more personalized and data-driven experience for students. As shown in the figure below, academic data assets that will be used to create personalized learning experiences include grades, social-emotional learning (SEL) insights, IEP (Individualized Education Plan) accommodations and restrictions, and teacher feedback. This data will be ingested into the student model and used to create recommendations curated for the user.

Grades, in particular, serve to contextualize each subject and align it with a corresponding RIASEC code, forming a foundational building block for the prototype. Over time, learning outcomes will be used as a benchmark to track students' career paths and progress. In the meantime, coefficients are applied to the academic data assets in order to create career recommendations that are aligned with the student.

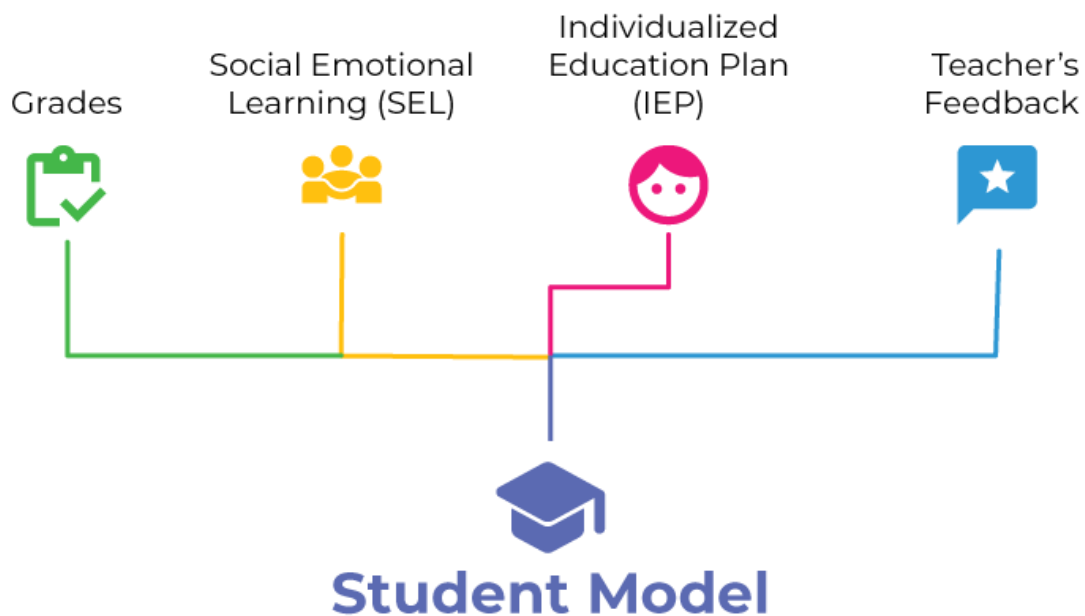


Figure 13 - Student Model - Academic Assets

To incorporate grades, coefficients were applied for each school subject, aligning them with each RIASEC code (Realistic, Investigative, Artistic, Social, Enterprising, Conventional). For each subject, a coefficient from 0 to 1 was assigned to indicate how closely the common core standard objective aligns with each RIASEC trait (skills and abilities). There was no prior research that applied this method directly to the common core, but scholars have matched MBA curricula to RIASEC work activities (K. R. Thompson & Koys, 2010), and mapped a plan for curriculum development and assessment to RIASEC (Freeman & Hirsch, 2008). While the exact approach has not been used, my research builds on these foundations by creating a grade-based alignment model that aligns the common core to RIASEC traits.

A scaling system was adopted in which a higher coefficient (closer to 1) indicates a stronger alignment; for example, an art class would have a coefficient of 1 for the "Artistic" trait. Each subject was evaluated to determine if it had no alignment (0), weak alignment (0.1–0.3), moderate alignment (0.5), or strong alignment (1.0). The RIASEC traits and alignment scale are shown in the figure below.

RIASEC-GRADE SCALE

RIASEC TRAITS	Activities/Skills Relationship to Academic Subject			
	Strong alignment (core activity of the subject)	Moderate alignment (related but secondary activity)	Weak alignment (minor or occasional activity)	No alignment (unrelated to the subject)
	1.00	0.50	0.10 - 0.30	0.00
Realistic (R): Hands-on, practical, physical tasks (e.g., sports, mechanics, lab work)				
Investigative (I): Analytical, research-driven, problem-solving (e.g., science, math, data analysis)				
Artistic (A): Creative, expressive, open-ended activities (e.g., arts, music, creative writing)				
Social (S): Interpersonal, collaborative, helping roles (e.g., teaching, group work, counseling)				
Enterprising (E): Leadership, entrepreneurship, persuasive tasks (e.g., business, debate, student council)				
Conventional (C): Structured, organized, detail-oriented work (e.g., coding, accounting, admin tasks)				

Figure 14- RIASEC-Academic Scale

The process of assigning a scale to each subject involved reviewing the educational Common Core standards for each subject listed in the figure below and evaluating how well these standards aligned with the career skills and abilities described in the O*Net content model. This method takes each student's grade and applies the coefficient for each RIASEC code. For instance, if a student's final grade in science is 80%, their RIASEC scores would be calculated as follows: R ($80 * 0.6 = 48$), I ($80 * 1 = 80$), A ($80 * 0.3 = 24$), S ($80 * 0.4 = 32$), E ($80 * 0.5 = 40$), C ($80 * 0.6 = 48$). These weighted scores are then applied to the student's interest profile.

RIASEC-GRADE COEFFICIENTS

Subject	R (Realistic)	I (Investigative)	A (Artistic)	S (Social)	E (Enterprising)	C (Conventional)
Math	0.4	1	0.2	0.3	0.4	0.9
Physical Education	1	0.5	0.1	0.6	0.3	0.2
STEM	0.5	1	0.3	0.4	0.5	0.6
Social Studies	0.3	0.5	0.4	1	0.8	0.5
Art	0.1	0.2	1	0.7	0.5	0.3
Business Studies	0.3	0.4	0.6	0.5	1	0.8
Computer Science	0.4	0.8	0.2	0.2	0.4	1
English	0.2	0.6	1	0.7	0.5	0.7
Science	0.6	1	0.3	0.4	0.5	0.6

Figure 15- RIASEC-Grade Coefficients

Grades are commonly used as indicators of academic performance and can be used in personalized learning models to tailor educational experiences to each student's strengths and needs. However, grades can also carry systemic biases that may disproportionately impact students from marginalized backgrounds, including those affected by socioeconomic disparities and racial biases. These biases may influence grades in ways that do not accurately reflect a student's true potential or learning needs. In contrast, a mechanism is required to accommodate students with Individualized Education Plans (IEPs).

According to an interview conducted with Myeisha Ramirez, Program Coordinator at Proviso Township High School District 209, students with IEP accommodations may have grading scales that differ from those used for students without IEPs. These grading scales are adjusted when accommodation is needed, which can result in modified student

grades. Additionally, Mr. Ramirez highlighted biases introduced by district policies, such as awarding points for simply ‘attending class’ or ‘turning in homework,’ which may not accurately reflect a student’s abilities. Another grading bias to consider is the ‘no-zero’ policy, where teachers are required to assign grades no lower than 50%, even if a student fails to submit assignments (Mhide et al., 2023). Normalization techniques are required to adjust grade data across different school settings, reducing the impact of variations in grading practices. This approach integrates grades alongside indicators in the “personal-fit” model to create a more balanced and comprehensive view of each student. This approach aims to provide equitable and culturally responsive personalized learning experiences, acknowledging that grades alone may not fully capture the capabilities of students from diverse backgrounds.

The social and emotional learning (SEL) data insights contribute to matching students to a RIASEC code using a similar methodology for grading and creating personalized journaling dialog. Similarly, the integration of teacher feedback into the model will be addressed in future phases, which are discussed in the Next Step section.

Cultural Opportunity Fit

The Cultural Fit Model integrates cultural identity, contextual relevance, and collaboration to create a more personalized and meaningful career exploration experience for African American boys. This model leverages culturally responsive elements by incorporating students' visual representation of identity through images, utilizing insights from their neighborhoods to inform career recommendations, and engaging them in co-design activities.

Cultural Identity

The O*Net’s IP method is adopted to customize the user experience (UX) to reflect the cultural identity of African American males. Drawing from prior research (Huang et al., 2022; S. E. Williams et al., 2024), the O*Net’s text-based survey has been replaced with an image-based version that features AAM in each survey question. Studies indicate that visual representation is part of a value system that, when aligned with identity, self-recognition, and cultural pride, reinforces a system of value (Lateef et al., 2024). In researching learning technologies that explicitly depict the cultural identity of the user, I found none. While this implementation may seem minor or overlooked in mainstream academic agendas, some scholars have researched this issue and believe that a strong sense of ethnic identity is crucial for young Black adults to support and protect their career aspirations (Lateef et al., 2024). According to (M.-M. Cheng et al., 2021), understanding a student’s culture helps teachers become receptive to diverse learning experiences and effectively communicate and receive information, which is also applicable when designing for students. This approach borrows from culturally responsive teaching (CRT), which is the use of cultural knowledge and past experiences to increase the efficacy of learning experiences for diverse learners. Culturally responsive design is a critical element in personalizing learning technologies because it aims to ensure that students' experiences and interactions with these technologies are relevant to their personal and academic interests and their cultural identities. This approach recognizes that technology cannot be effective unless it is tailored to the cultural contexts of its users (Sawyer, 2022), particularly marginalized students like African American boys, who face systemic barriers in education and career exploration. As highlighted by (Ogbonnaya-Ogburu et al., 2020), the field of Human-Computer Interaction (HCI) has long recognized the importance of adapting technology experiences to align with users’ cultural backgrounds. Technology that is personally relevant promotes positive engagement and creates an environment where

students feel represented and valued. This approach enhances user motivation and overall interaction with the learning tools.

Contextual Relevance

Cultural responsiveness also extends to the way technology is contextualized within specific learning environments. As (Egede et al., 2024) suggest, contextualizing culturally aware technology is a key aspect of inclusive design. Embedding cultural context means not only considering how technology interacts with cultural identities but also how it can address the unique challenges faced by underrepresented students. One such challenge is the underrepresentation in certain industries: manufacturing wholesale (1.4%), information (1.6%), and construction (2.9%), to name a few. Using statistics from the American Community Survey (ACS), which gathers data on population, housing, economic, educational, and employment information across the United States, this model leverages ACS's data to assign high coefficients to careers where African American males are underrepresented. Students are encouraged to explore these careers, with the potential of raising awareness and, hopefully, contributing to greater equity if they choose to pursue these paths. By focusing on careers where African American males are historically underrepresented, the model aims to broaden students' perspectives and empower them to envision possibilities beyond traditional stereotypes. Furthermore, the use of neighborhood-specific data serves a dual purpose: tailoring career exploration activities to be either virtual or in-person based on accessibility and relevance. This approach ensures that the career exploration process is not only culturally resonant but also practically aligned with the students' lived experiences and local opportunities.

Collaboration

In the context of this research, cultural responsiveness goes beyond simply acknowledging cultural differences—it demands that those directly impacted by the technology be active participants in its design. Collaborating with users aligns with the work of (Drame & Irby, 2016; Egede et al., 2024), which stresses that culturally tailored technology must be led by individuals who are directly affected by its use. This manifests through the co-design process, where students contribute their lived experiences to the development of learning pathways, ensuring that the resulting tools are culturally aligned and reflective of their needs. (Egede et al., 2024) noted that understanding what it means to center underrepresented individuals from the perspective of Black creators and technologists is vital to achieving a more inclusive and effective design process. By incorporating these principles, this research aims to create learning technologies that not only adapt to students' academic and career needs but also reflect and honor their cultural experiences. Discussed later in the Next Step section, the plan of working with African American males will be discussed in more detail.

Personalized-Fit

Two things were done to personalize and ensure these classifications are tailored to the developmental stage of the targeted audience.

First, building off the O*Net content model discussed in Section 2.3, students will complete a 30-question survey that provides their RIASE profile. That profile will be leveraged to incorporate the interests and hobbies that the students provided. This approach factors in careers and hobbies that may not be listed in the survey, and it helps build a stronger student profile based on the individual student. It also builds off Timnit Gerbu's theory of building a curated data model that is aligned with particular users and does not utilize a large model that may incorporate biases (Bender et al., 2021). The goal is to collect longitudinal data that captures the evolving interests and hobbies of African American boys.

Secondly, while not novel, age-appropriate categories were used in my prototype to help simplify the categories and make the naming convention easier to understand; instead of RIASEC, BTCHLP (builders, thinkers, creators, helpers, leaders, and planners) are incorporated into the prototype as illustrated in the figure below. This method of adjusting language and content to suit the cognitive and emotional maturity of the students is a key aspect of this research to personalize the learning experience, making it more relatable and accessible for each individual.



Figure 16 - RIASEC Cultural Relevant Hexagon

Experiential Learning – (Adaptability)

Educators and instructional designers must leverage experiential learning theory so that students can fully engage and extract meaningful learning from rich educational experiences. A core tenet of experiential learning is that "learning results from synergetic transactions between the person and the environment" (Kolb & Kolb, 2009). Just as a river shapes the landscape through its continuous flow, experiences shape the learner's understanding through ongoing interactions with their environment. This interaction facilitates the transformation of experience into knowledge through a structured process known as the experiential learning cycle (Kolb & Kolb, 2009).

Experiential learning is a dynamic, continuous process of meaning-making within specific contexts, where students develop an interest and perceive the relevance of learning through personal and environmental interactions. Students will engage with personalized, interest-based recommendations that provide opportunities for hands-on exploration through both formal and informal activities. These experiences allow students to participate actively in the Concrete Experience (CE) stage of Kolb's cycle, fostering real-world engagement.

For instance, each career path recommendation is accompanied by activities that help deepen students' understanding of the profession. These activities may include watching documentaries, reading articles, interviewing professionals, or discussing career opportunities with teachers. Importantly, these tasks are aligned with the career path and tailored to the student's geographical and cultural context, ensuring relevance and practicality.

Take the example of a student interested in a science-related career aligned with the "Investigative" RIASEC type. This student might be tasked with the following:

- Watching a documentary on scientific breakthroughs,
- Reading an article about recent discoveries in their field,
- Discussing career options with their science teacher.

After participating in these activities (Concrete Experience), students enter the Reflective Observation (RO) phase, analyzing their experiences and reflecting on what they have learned. This reflection informs the next stage, Abstract Conceptualization (AC), where students formulate new ideas or revise their understanding of potential career paths. The process culminates in Active Experimentation (AE), where students test their newfound ideas through practical actions, such as volunteering, internships, or informed academic decisions.

This cyclical process encourages continuous interaction with learning technology, where students provide feedback on their experiences. The feedback loop allows technology to further adapt to the student's evolving interests and behaviors, ensuring that the experience remains personalized and relevant. The students can be empowered to take actionable steps toward achieving their goals as they gain deeper insights into their career paths and real-world challenges. According to Learning Science, effective learning environments monitor a student's knowledge development through a learning trajectory rather than solely relying on standardized testing (Sawyer, 2022). An approach that can achieve this is a career learning trajectory map that records the activities students engage in and captures their feedback to assess alignment with career activity objectives. This approach helps track the student's learning path and displays their progress for them to view and understand.

By embedding experiential learning theory into the design of learning technologies, educators and instructional designers can create authentic, hands-on learning experiences that encourage active participation, reflection, and real-world application. Emerging technologies can further enhance this process by integrating real-time data, enabling continuous adaptation of learning paths to meet students' evolving needs.

Adaptability Tool

Building on the collaboration between learning technologies and students, it is essential to recognize the role of tools that enable adaptability. One such tool is Artificial Intelligence (AI), which provides the capacity to analyze academic data and student profiles, learn from interactions, and make real-time decisions to adjust the learning experience. Career curricular design involving marginalized communities (Compton, 2021; McFarlane-Beau, 2021), and technology design (Han et al., 2022; D. P. Williams, 2021) has demonstrated that valuable insights derived from observing user behaviors in real-world contexts can guide designers' understanding of interactions and inform the development of future technologies.

Artificial Intelligence (AI), specifically machine learning (ML), develops algorithms and models to learn from data and perform specific tasks without explicit programming (Shelehov et al., 2023). ML plays a critical role in enhancing the personalization and adaptability of learning technologies and can analyze student interactions, including responses, language patterns, and engagement with content, to provide real-time feedback and adjustments to career exploration pathways. This type of tool enables the system to suggest learning paths that are both culturally relevant and aligned with students' career interests, ensuring that the content they engage with reflects their lived experiences and aspirations. For instance, when students are recommended resources and activities to explore and learn about a career, technology can prioritize virtual and in-person activities based on a child's geographical area and the resources available to the student. Resource-driven filters create a more effective learning experience tailored to the student's local resources.

Ethical AI design (Taddeo et al., 2024) is integral to ensure that the learning technologies are designed with fairness, transparency, and privacy in mind. Safeguarding student data, particularly sensitive information like SEL and IEP data, is critical to maintaining trust between students, educators, and parents. By incorporating culturally responsive teaching methods, experiential learning, and aligning adaptive experiences with IEPs, the research aims to provide educators with tools that enable them to create more inclusive and responsive educational experiences for marginalized students. These learning technologies can help teachers recognize and overcome potential biases by exposing both students and educators to career paths that students may not have considered. In doing so, the tools broaden educators' understanding of the diverse opportunities available to their marginalized students, enabling them to create an environment that encourages engagement, increases career awareness, and simplifies complex career trajectories. According to the learning science communities, technology should facilitate helping learners experience deep learning and represent abstract knowledge in a more concrete form (Sawyer, 2022). Ultimately, this empowers educators to better support their students' long-term success through personalized and adaptive learning pathways.

Participatory Design (Co-design)

Participatory design is a collaborative approach that has traditionally focused on bringing awareness to social justice and influencing future work. Co-design, an evolution of participatory design, takes this concept further by emphasizing the active collaboration of multiple stakeholders throughout the design process. (Potvin et al., 2024). While participatory design is often characterized by engaging limited stakeholders in predefined design outcomes,

contemporary research highlights a shift toward deeper, long-term engagement with diverse communities and broader social and technological ecosystems (Bang & Vossoughi, 2016).

In my research, co-design builds on the foundations of participatory design but advances it through a collaborative process that includes students, teachers and administrators, parents, and design students. This collaboration is central to identifying and addressing cultural gaps often overlooked in traditional learning systems. By actively involving these stakeholders, the co-design process promotes a stronger connection to the material and builds stronger relationships with the stakeholders (Potvin et al., 2021).

Co-design, which literally means "collaborative design," focuses on engaging all individuals directly involved in an issue, place, or process in the design and sometimes even in its implementation (Burkett, 2012). This approach brings co-creating learning technologies that are both innovative and impactful and leverage the collective experiences, skills, and knowledge of all stakeholders. By building mutual understanding across the entire educational ecosystem, co-design moves beyond just including the voices of end users—it facilitates a comprehensive engagement that incorporates the perspectives of all contributors, ultimately driving more meaningful and sustainable solutions to complex social challenges (Potvin et al., 2024). The prototype phase, detailed in later sections, will demonstrate that co-design methods can effectively build relationships (Paay et al., 2023) with African American boys by nurturing trust and granting them agency (Jackson et al., 2022) in the research process. This approach ensures that they are seen as the experts in the room, emphasizing attentive listening and the incorporation of their feedback, making the prototype truly reflective of their experiences (Drame & Irby, 2016; Egede et al., 2024).

3.2 Data Collection and Analysis

This research employs a mixed-methods approach, combining quantitative and qualitative data collection techniques to provide a comprehensive evaluation of how personalized learning, co-design, and experiential learning frameworks impact African American boys in marginalized communities. The data collection tools used include surveys, observations, interviews, and workshops, allowing for a multi-faceted understanding of student engagement and the effectiveness of the interventions.

Surveys: Both pre-assessments and post-assessment surveys will be administered to evaluate career awareness, interest, and understanding. The pre-assessment survey establishes a baseline of the participants' knowledge and aspirations. The post-assessment survey assesses the workshop's impact, including how the student's career aspirations, interests in particular fields, and engagement with learning materials have evolved after participating in the intervention.

Observation: During the workshops, observations will be conducted to capture real-time student engagement with the prototype and learning tools. This method will focus on measuring how students interact with culturally relevant content, identifying patterns in their behavior, and understanding how they navigate the learning activities provided.

Interviews: A random selection of students will be chosen to participate in one-on-one interviews. These interviews will provide deeper insights into the students' experiences, focusing on how the learning technologies and interventions influenced their career awareness, aspirations, and engagement with culturally relevant content. The

questions asked during the interviews will be derived from the pre-assessment and post-assessment surveys to ensure continuity and to explore in greater detail the themes and patterns that emerge from the survey data. The interviews will also allow for a more nuanced understanding of the co-design process and the perceived effectiveness of personalized recommendations.

Workshops: A key component of the data collection process involves interactive workshops, where students actively participate in character creation, a process designed to align with their career interests. This experiential learning method provides insights into how well the co-design and personalized learning elements are functioning, allowing students to voice feedback on their experiences.

Measuring the Impact of Cultural Relevance

One of the key challenges of this research is effectively measuring the impact of cultural relevance in learning technologies. Cultural relevance is designed to resonate with the specific lived experiences of African American boys. Clear metrics have been defined to evaluate the effectiveness of the interventions. These include:

Student Engagement: Tracking how students interact with culturally tailored content during workshops and class activities.

Learning Outcomes: Measure career awareness, aspirations, and understanding of various career paths post-intervention.

Feedback Loops: Gathering data through post-assessment surveys and interviews to assess how students perceive the cultural relevance of the content.

Outcome Evaluation

The success of this research hinges on evaluating the effectiveness of each framework. For personalized learning, success will be measured by how well the personalized recommendations align with the student's interests and goals. For co-design, the focus will be on assessing how students' and stakeholders' input was integrated into the learning experience and how cultural relevance influenced student engagement. Experiential learning outcomes will be measured by the practical application of learning, such as the student's ability to engage in real-world career exploration activities and provide continuous feedback on their learning experiences.

3.3 Limitations and Ethical Considerations

In the realm of prototype-based research, especially when dealing with sensitive topics, the concept of data governance is pivotal. It encompasses the comprehensive management of data availability, usability, integrity, and security. The importance of data governance lies in its role in safeguarding personal and sensitive information, which is often at the core of such research. By implementing stringent policies for data privacy and security, the platform ensures compliance with legal standards like GDPR, HIPAA, or FERPA, as well as ethical norms. This includes clear privacy policies, robust security measures to prevent unauthorized access, and adherence to legal and

ethical data usage. Effective data governance is key in building trust with participants and ensuring that their sensitive data is handled with the utmost responsibility and care.

In the context of AI-driven learning technologies, the systems must account for data security and the potential ethical challenges related to parental access to sensitive student data. By developing transparency in the design and decision-making processes, it becomes possible to balance the ethical considerations across stakeholders, ensuring that the system respects both student autonomy and parental involvement. Independent validation of these systems ensures that the system's trustworthiness is aligned with the expectations of all stakeholders involved, particularly students and their parents. This approach encourages both academic and career development while addressing critical concerns around information sharing and data security.

One key limitation of this research is the inability to gather longitudinal data. While the personalized learning framework dynamically adapts to student needs, interests, and aspirations in real time, the research lacks the capability to track students' long-term progress. Without longitudinal data, it is challenging to fully assess the lasting impact of adaptive learning technologies on career exploration and academic growth. This limitation suggests that future studies should focus on longitudinal research to understand the long-term outcomes of personalized learning, co-design, and experiential learning methodologies.

Another limitation involves scalability. Although the co-design process empowers students and personalizes learning experiences, implementing this model at scale in resource-constrained environments, particularly in marginalized communities, may prove challenging. The time and effort required for continuous feedback loops and active participation from students, teachers, and parents could become resource-intensive. Teachers in these communities often lack the time and technological resources to monitor and adjust learning pathways based on real-time feedback continuously. Future research and practical implementations should explore automated systems that provide timely notifications to educators when significant changes in a student's learning path occur, reducing manual oversight.

Ethical Considerations

The use of AI within the personalized learning framework presents several ethical concerns that must be addressed to ensure the integrity and fairness of the learning experience. These include:

Data Privacy: The collection and use of personal student data—such as academic profiles, behavioral feedback, and cultural information—raises concerns about the security and confidentiality of sensitive information. Students and parents must trust that their data will be handled responsibly. Data privacy protocols are suggested to ensure compliance with legal and ethical standards, including anonymization of student data and transparency regarding how the data is used.

Algorithmic Bias: AI-driven personalized learning systems may inadvertently perpetuate biases in their recommendations, particularly for marginalized communities like African American boys. To address this, I will implement mechanisms to continuously audit and evaluate the AI's decision-making processes, ensuring that culturally relevant content is incorporated and that any biases are identified and corrected. The inclusion of multiple stakeholders

in the co-design process is also intended to mitigate biases, as students, teachers, and parents will provide diverse perspectives and input into the system's development.

Over-reliance on Technology: While AI provides valuable insights and adaptability, there is a risk of over-relying on technology, potentially diminishing the role of educators in shaping the learning process. To counter this, the design of the learning framework emphasizes co-collaboration between technology and educators. Teachers and administrators play a key role in refining the learning pathways, guiding students, and providing human oversight where technology may fall short.

Student Agency and Control: Empowering students while maintaining ethical boundaries regarding the visibility of their personal information is critical. Students must have agency over what data is shared with their teachers and parents, particularly in terms of career aspirations and personal progress. Ethical considerations will include safeguarding student agencies by allowing them to control which insights are shared and ensuring their active participation in shaping their own learning experiences.

Addressing these ethical concerns is crucial to establishing trust in the use of AI and adaptive learning systems. Ethical safeguards will be embedded throughout the research process to ensure fairness, protect data privacy, and avoid biases in recommendations. Additionally, to overcome these challenges, continuous collaboration with teachers, administrators, and students is necessary to align technological advances with human-centered education values.

4 Preliminary Findings and Simulated Data

4.1 Preliminary Findings

Case Study: Equity and Immersion: Strategies for Creating an Inclusive and Engaging Educational Environment

This qualitative case study, entitled "Equity and Immersion: Strategies for Creating an Inclusive and Engaging Educational Environment," explored the efficacy of a career exploration workshop tailored for students from marginalized communities. The primary goal was to investigate how design-driven career activities could impact career awareness among these students, aiming to promote an inclusive and equitable educational experience. This objective was achieved by enabling participants to investigate possible career avenues by developing websites that projected their future professional selves. Additionally, the secondary aim was to evaluate the level of engagement within the workshop, its perceived value to the participants, and the practicality of the skills acquired for future application.

4.1.1 Method

An exploratory qualitative design was used to shed light on how a phenomenon manifests by allowing participants to contribute to the generation of new ideas and hypotheses regarding the phenomenon of interest (Foster, 2024). This design allows the use of group discussions to unpack lived experiences, clarify concepts, and formulate research problems (Jelonek & Urbaniec, 2019). This methodology was instrumental in investigating the career interests of high school students through a career workshop, which encouraged participants to envision their future careers by designing websites based on their career aspirations. This approach not only shed light on the impact of career-focused workshops in raising awareness among participants from underserved communities but also created a forum for career discussions. Incorporating surveys into this methodology allowed for the collection of participants' insights on the advantages of the workshop, thereby enhancing the depth of the study's results. Future research into the effectiveness of such workshops in assisting high school students to envision their future professional identities is needed.

The research was conducted during a summer workshop facilitated by TechCo Savy, a nonprofit organization on the west side of Chicago, Illinois, that provides STEM workforce skills to African American youth, the majority of whom come from Chicago Public Schools ("CPS"). CPS is ranked as the third-largest school district in the United States, and it has a history filled with both significant achievements and formidable challenges (Guo-Brennan, 2020). Throughout its history, CPS has grappled with ongoing issues resulting in unequal distribution of resources, favoring schools in wealthier neighborhoods over those in less advantaged areas. The classroom environment, shaped by staffing difficulties and historical contexts, exacerbates the challenge of providing all students with a high-quality education.

The backdrop of racial segregation, economic disparities, and political influences has profoundly influenced CPS's history. Current statistics show a student body primarily comprised of African American (35%) and Hispanic (46%)

students, with a significant academic performance gap evident when compared to their White counterparts. This gap underscores the educational disparities rooted in socioeconomic status, access to quality education, and a scarcity of motivational figures. These deep-seated challenges highlight the critical need for innovative approaches that go beyond traditional educational methods. There is a pressing demand for new strategies, tools, and platforms to address these issues effectively and comprehensively.

In light of these challenges, there is a clear and urgent need for workforce skills development among CPS students. Equipping students with the necessary skills for the workforce is essential in bridging the achievement gap and ensuring that all students, regardless of their socioeconomic background or race, have equal opportunities for success in their future careers. Innovative solutions, including the integration of workforce skills training into the curriculum, are vital in preparing students for the demands of the modern job market and in fostering a more equitable and inclusive educational environment.

4.1.2 Participants

Recruitment of participants occurred during a summer workshop organized by TechCo Savvy. Twenty-five participants attended the summer workshop, and twenty elected to participate in the research. Eleven students were in attendance to complete the actual survey, with two identifying as female and nine as male. Among these participants, ten identified as African American and one as Hispanic. All participants were students at a CPS high school, aged between 15 to 17 years old.

4.1.3 Study Approval

Authorization for this research was granted by the Institutional Review Board (IRB) at the Illinois Institute of Technology (IRB-2023-87). Given that all participants were minors under the age of 18, they filled out an assent form, while their parents provided consent through a parental consent form. The ethical protocols adhered to in this study align with the guidelines set forth by the IRB concerning research involving human subjects.

4.1.4 Activity Observation

In this career design workshop aimed at validating the impact of design-driven activities on career awareness for marginalized students, participants were encouraged to articulate their career ambitions. The narrative of the workshop centered around me taking on the role of a financial backer, with the stipulation that participants establish an online business presence. The three-week workshop was not intended to impart coding skills but rather to actualize participants' career ideas. To achieve this, Wix.com was chosen for its ease of use, where each participant selected a template that resonated with their business type, either service-based or retail.

Participants 19 and 7, possibly close friends, decided to collaborate. Participant 19, who already had a clear business vision, expressed her preference for simplicity over technical complexity, stating, "I attended classes on website creation, but they focused too much on coding. I am not interested in coding; I just want to showcase my creations." Her enthusiasm motivated her to develop her website content at home further. She was driven by excitement, and she explained, "I was so excited that I could create a website that I could use to make money." Participant 19 demonstrated

self-determination skills vital for her autonomy but lacked the knowledge on how to execute her plan. A culturally responsive approach underscores the importance of engaging community-based organizations, as highlighted in the article "It Takes a Village: Understanding African American High School Students' Self-Determination in School," which can provide the necessary support and opportunities for African American students to develop and apply their self-determination skills in meaningful ways (Janise S. Parker & Sarathy, 2020).

Similarly, participant 11 pursued an entrepreneurial path in fashion, aiming for financial independence: "I want to sell my clothes to make money, so I do not have to depend on my mother." The need for financial independence reflects a broader narrative among students from socio-economically disadvantaged backgrounds who view entrepreneurship as a pathway to financial autonomy and family support, as documented by (Johnson et al., 2021). These students often view entrepreneurship not just as a career choice but as a means of achieving financial independence and contributing to their families' well-being. This highlights the critical role of such workshops in empowering students in marginalized communities with the skills and confidence needed to pursue their aspirations. Allowing students to explore the idea of their future career selves is at the forefront of this research, as participant 15 explained, "I never thought about starting a business but seeing her (*referring to participant 15*), I think I can create me a business. I like to create flyers for people at my church".

Participant 1 initially expressed skepticism in the workshop, stating, "I do not find it useful for what I want to do." When asked to elaborate, he disclosed, "I am an artist. I draw animals and futuristic art." Further inquiry into his skepticism about the usefulness of a website for an artist revealed underlying insecurities. He explained, "No one will buy this type of art. I only do this for fun," indicating a common apprehension among students about their creative endeavors (Vaughn, 2023). In addition to the lack of confidence that Participant 1 exhibited, he was further distracted by the act of creating the website and expressed, "I would hire someone to do this for me. Why do I need to learn how to create a website." This comment ignited conversation among the participants, and participants (7, 11, 13, 19) echoed the importance of learning how to operate your business and not depend on other people. The conversation among the participants enriched the research and demonstrated the importance of relating content to real-world situations in the learning process, as well as the impact of learning from peers. After a debate on the issue, Participant 1 remained firm in his belief. However, Participants 9 and 18, who were initially browsing the internet and resting on the desk, respectively, became involved in the discussion and attempted to begin work on their website, though they ultimately ran out of time.

This workshop not only showcased the entrepreneurial potential of African American students but also emphasized the critical role of educational initiatives that provide a platform for these students to envision and take practical steps toward their career goals. The workshop demonstrated the transformative power of education in bridging the gap between aspiration and reality, allowing students to explore and affirm their future career identities. Moreover, insights gathered from such workshops can directly inform the design of AI-driven career learning platforms, ensuring they are equipped with features that reflect and support the real-world needs and preferences of users. These experiences, when integrated into AI algorithms, can enhance the platform's ability to offer personalized guidance, making it a more effective tool for career planning and development.

4.1.5 Data Analysis

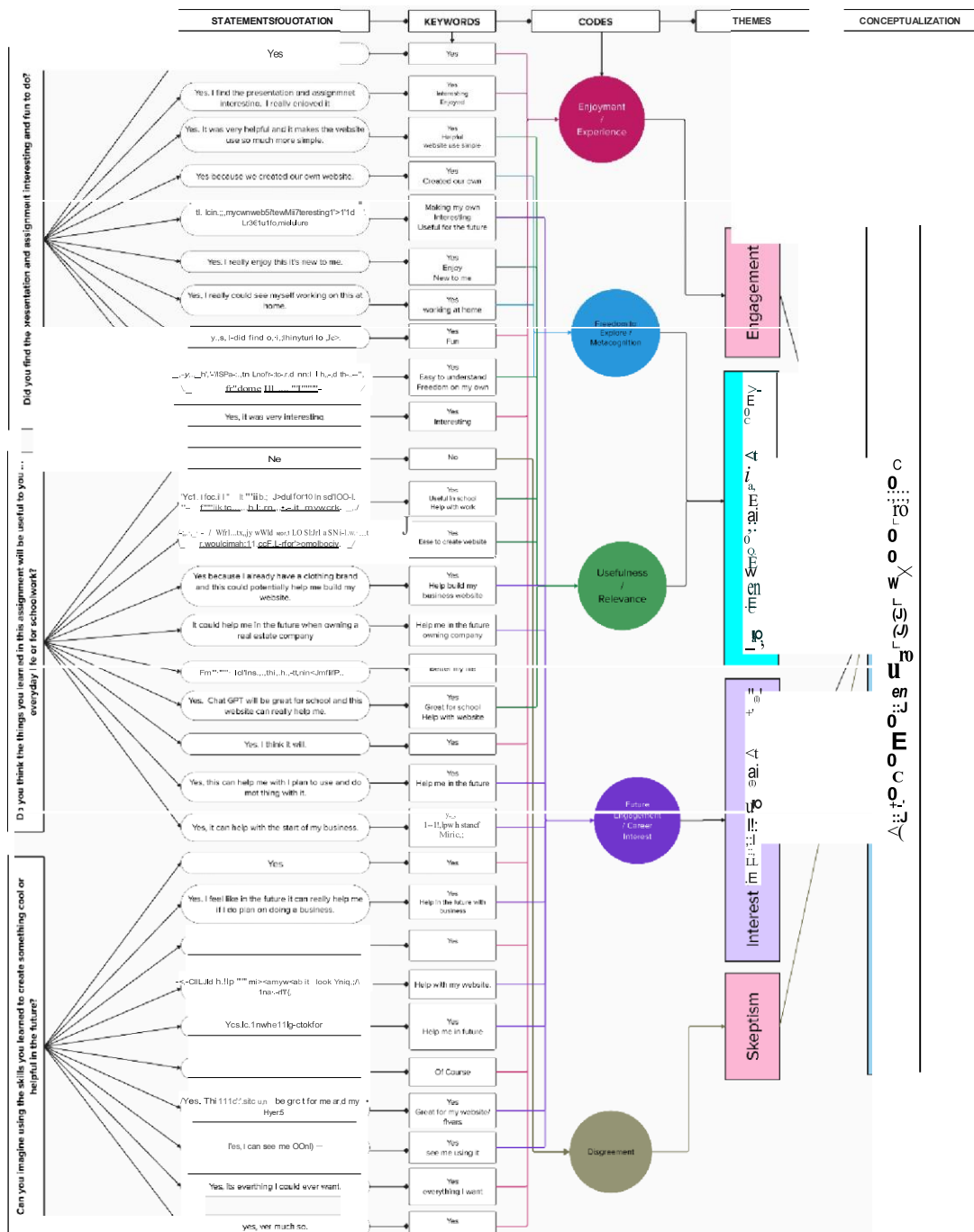
Following the workshop, participants were requested to fill out a survey that included demographic details and three questions. The purpose of the questions was to establish the engagement level, the perceived usefulness of the workshop, and the need for future application.

As seen in Figure 2, the data from the surveys were analyzed using a systematic approach for thematic analysis by (Naeem et al., 2023). Systematic thematic analysis (STA) was used to establish a method of qualitative data analysis that is concerned with the examination of shared meaning in relation to a topic or a phenomenon. STA is a six-step process approach that allows coding and a nuanced reading and sense-making of the data (Naeem et al., 2023). These theoretical underpinnings of systematic thematic analysis make this method well-suited for the analysis of the present data. STA is conducted in six phases: transcribing and familiarizing yourself with the data, identifying keywords, generating initial codes, developing themes, creating concepts from the themes, and creating a conceptual framework. This analytic approach is anchored in ground theory and expanded through the addition and interpretation of concepts and conceptual frameworks. Before iterating through the STA process, the participant's identification was removed, and each participant was provided with a unique identifier.

During the statement/quotation phase, each participant's quote was logged into a table as it was written. The information was carefully reviewed to get a familiarization of the content provided and to identify any patterns; common wording was color coded for future reference. The second phase is the coding phase, which identifies important concepts, reoccurring patterns or terms, or keywords that relate to overall research. During the third step, codes are assigned to the data that represents any themes relevant to the research process, and similar and dissimilar key concepts identified across the quotes are categorized into themes. In the concept phase, the themes were named and interpreted.

This qualitative case study explored the effectiveness of a career exploration workshop designed for students from marginalized communities and revealed the significant role of educational initiatives in helping students envision and take practical steps toward their career goals. This research emphasized the importance of experiential opportunities that allow students to explore and affirm their future career identities and that will directly inform the design of learning technologies.

Student's Emotional Relationship to Their Future Career Selves



Ncietem, M, Ozuem, W., Howdl, K., ...: Ranf Agnl, S. (2023). A Step-by-Step Process of Th'm1tr1c An11yl1s1s /o Del1'op fl Concep1fl M Model in Qu1l1rtat1ve Res&mch. International Journ1 fl ot Oufll1tat1ve Melhod5, 22. <https://doi.org/10.1177/16094069231205789>

Figure 17- Student's Emotional Relationship to Their Future Career Selves

4.2 Synthetic and Simulated Data for Prototype Validation

Conducting research with marginalized communities requires sensitivity, particularly within the African American community (Drame & Irby, 2016). Prior research has contributed to a level of mistrust, so researchers should approach the co-design process with caution (Egede et al., 2024). Including factors such as age and gender adds complexity, especially as children bring a certain type of affordance or naivety to the process. These vulnerabilities make career suggestion research even more intricate, requiring careful consideration of children's unique developmental perspectives and ensuring recommendations align with their emerging identities and aspirations.

The preliminary findings mentioned above support a career-focused approach, but before testing the prototype directly with African American male students, a combination of synthetic and simulated data methods was utilized to validate the prototype's foundational elements. Synthetic data was generated using a Python library (Fössing & Drechsler, 2024), which created 10,000 student profiles. Each synthetic profile included simulated responses to an interest-profile (IP) survey, resulting in RIASEC codes. Academic grades for math, science, and social studies were sourced from BSICS and anonymized, with identifying information removed. From the larger dataset of synthetic students, 200 student profiles were derived from real academic data, which were then incorporated into the synthetic dataset.

The decision to generate 10,000 synthetic profiles despite having only 200 real academic data profiles was a strategic methodological choice. Creating a larger synthetic dataset ensures the scalability and flexibility necessary for iterative testing (Walonoski et al., 2018). This approach allows for future testing phases to be conducted efficiently without requiring the continuous collection or generation of new data, which can be time-consuming and resource-intensive. Synthetic data provides a diverse range of profiles, enabling exploratory analysis and ensuring that the prototype can handle varied scenarios and student characteristics, making it robust for real-world applications.

In the simulated data process, 200 students were processed through a simulation model to generate interest profile (IP) and occupational-interest profile (OIP) matches. For each simulated match, three profiles were generated: occupational-interest profile fit (OIP fit), cultural fit, and academic fit. Simulated data methods enable early prototype testing under varied conditions, offering a reliable way to validate key functions and insights before direct empirical testing (Lehtinen & Raerinne, 2023). This approach helps to confirm and refine my findings, ensuring the prototype's stability and suitability before applying it in a live setting.

Simulations reveal patterns and test the model's basic functions (Strobl & Leisch, 2024). According to (Peschard, 2011), "epistemic cooperation" emphasizes that simulations do not replace experiments entirely but are critical in preparing robust prototypes for empirical testing, particularly when working with sensitive populations like students. Thus, after completing the simulation phase, the next step will involve empirical testing with African American boys, as detailed in the "Next Steps" section.

Careers were drawn from the ACS to focus on fields where African American males are historically underrepresented, with high coefficients assigned to these careers during the matching process. The larger synthetic dataset and the simulated data analysis allow for iterative testing, ensuring that future research phases can be conducted efficiently. This phased approach not only validates the prototype's foundational elements but also ensures that it aligns with the aspirations and needs of the target audience, addressing systemic barriers in career pathways.

5 Prototype Design and Future Directions in Research

5.1 Prototype

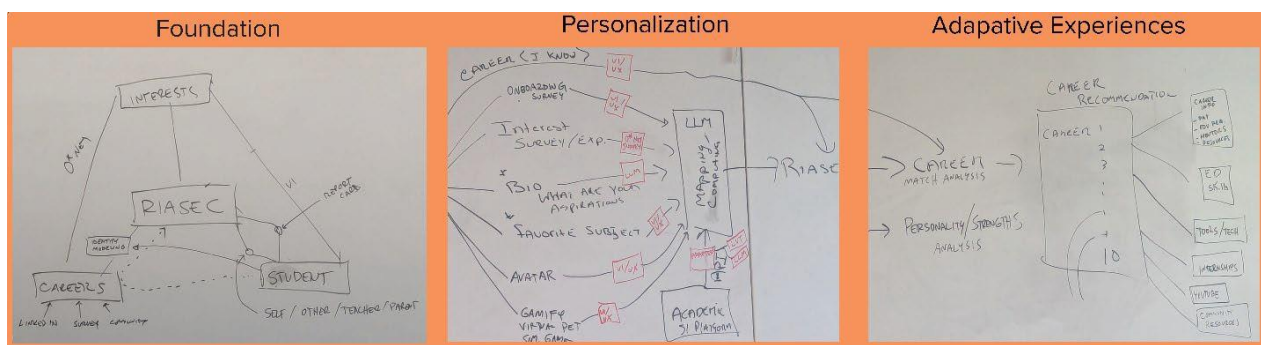
Design Students (Master Students from the Design Community):

This research serves as a model for how the design community can engage in collaborative, cross-disciplinary work that has real-world applications and makes a difference for underserved populations. In a 12-week collaboration with design students, who are master's students contributing to the design of learning technologies, created a unique space field with an exchange of ideas and perspectives that led to the development of inclusive design solutions. This collaboration grasped the importance of co-design, which prioritizes the voices and needs of the students (Costanza-Chock, 2020), specifically the needs of African American boys. It provided an opportunity for design students to understand the broader struggles AAM often faces. As well as what Black technologists face in the design industry, including the challenge of developing culturally relevant tools for their community. As (Egede et al., 2024) state in their article For Us By Us: Intentionally Designing Technology for Lived Black Experiences, *"it is important to understand how Black technologists design and develop tools and experiences for their community and every day."* This collaborative effort and its implications for addressing bias and cultural relevance in design will be discussed in detail later.

Prototype Design

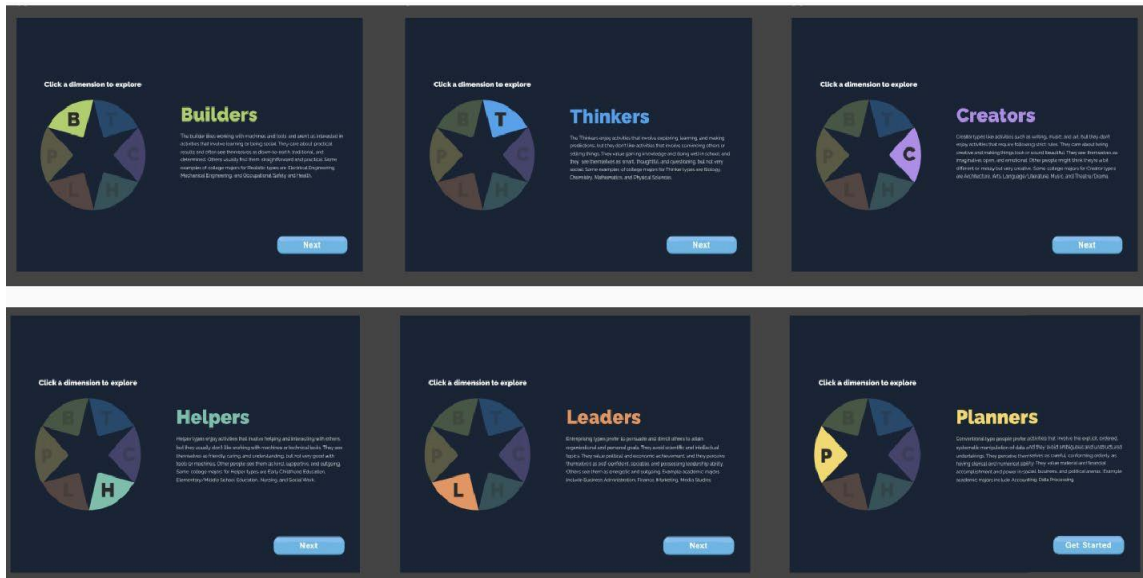
The prototype, developed through this research and named NorthStar, leverages the O*Net model, interest-based theory, academic data, and detailed student profiles to create a personalized career-focused tool. NorthStar enhances the connection between academic achievements and career planning (Zong, 2023; Romanov, 2023). This approach lays the foundation for a transformative impact on career guidance, with the potential to reshape how educational accomplishments lead to real-world success

As the user experience (UX) was refined, as seen in the figure below, the focus remained on creating a highly personalized and adaptive journey. Key features integrated into the design process included the personalization of the student's experience and the adaptive components for both informal and formal learning.

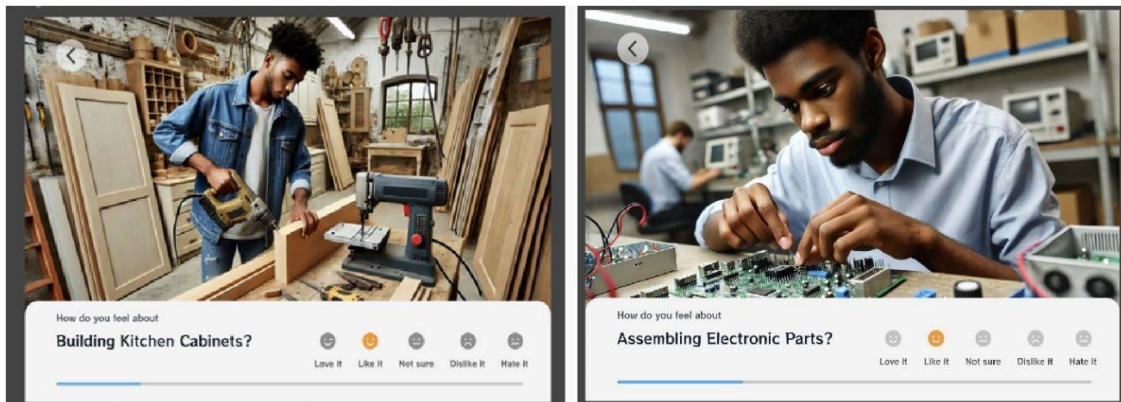


During the iterative design process, age-appropriate and culturally reflective elements were incorporated to address biases and enhance student engagement by offering a personalized and intuitive career exploration experience. This

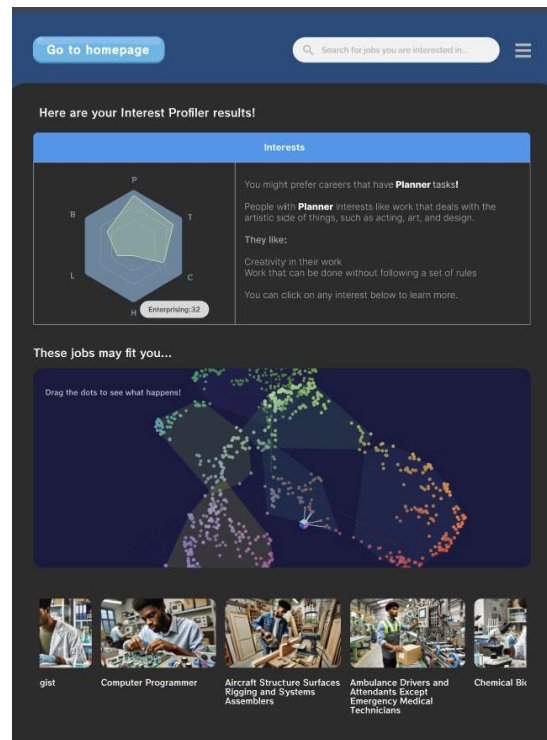
stage of the process tackled several complex biases, including the absence of culturally reflective language and media, which, as previously mentioned, can significantly hinder engagement and learning for marginalized audiences. To further personalize the user experience, we specifically adopted age-appropriate categories and descriptions for the RIASEC codes. Each descriptor is visually represented in a way that is simple enough for middle school students to understand. In addition to the categories, a brief, clear, and age-appropriate description of what each RIASEC code represents is provided, ensuring students can easily grasp the essence of each career path. The design, as shown in the figure below, incorporates color schemes that are age-appropriate and culturally reflective.



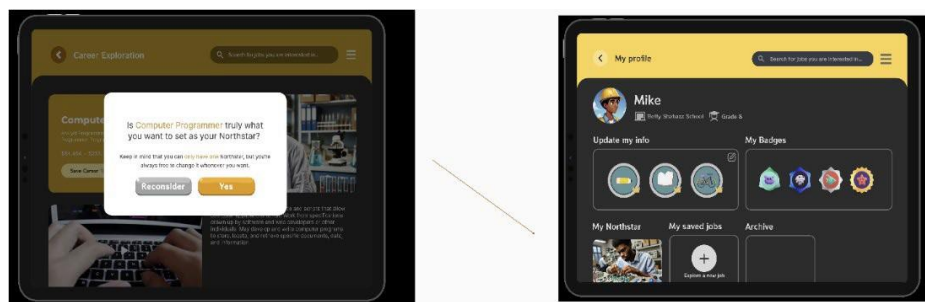
As mentioned earlier, culturally reflective images of African American boys were used in the prototype. Prompt engineering was used to generate images of African American boys representing the various careers across the occupations listed in the O*Net Database. As seen in the figure below, each image was accompanied by five emojis designed to capture different emotions or reactions from the participants. These emojis are aligned with research done by (Rounds et al., n.d.), to ensure that there are no loss in psychometric information by using emojis. The inclusion of visual elements like emojis and images made the survey less daunting and more enjoyable, moving away from the traditional, extensive, and often boring survey formats typically used with children.



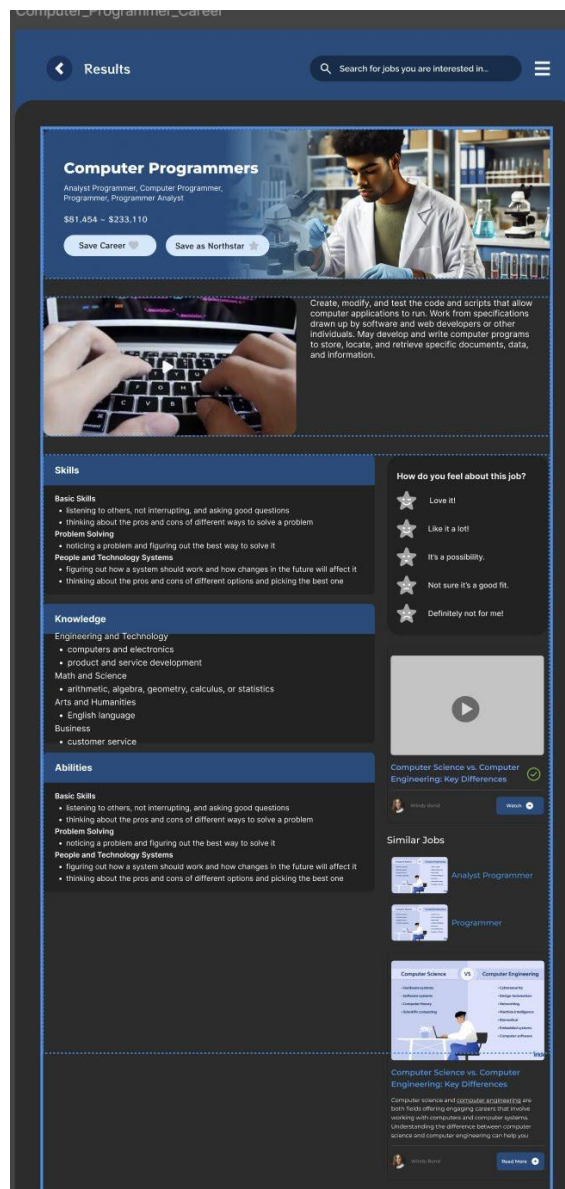
Career exploration provides students with critical exposure to unfamiliar career options by offering detailed insights into the career. In this phase of the prototype, once the student completes the survey, they are presented with a list of careers aligned with their top three RIASEC categories. During the exploration, as depicted in the figure below, students are given a detailed breakdown of each career, including the required skills, knowledge, and abilities, along with an accompanying video that offers an inside look into what it means to work in that profession.



For instance, if a student named Mike receives a recommendation for the role of a computer programmer, Mike can watch a video that illustrates a day in the life of a computer programmer while also exploring the specific skills and competencies necessary for the job. If Mike finds this career appealing, he can mark it as his "NorthStar", saving it as a personal career goal, as shown in the image below. Once Mike selects his NorthStar career, such as computer programming, his avatar will be updated to reflect that choice, visually aligning his virtual identity with his career aspiration.



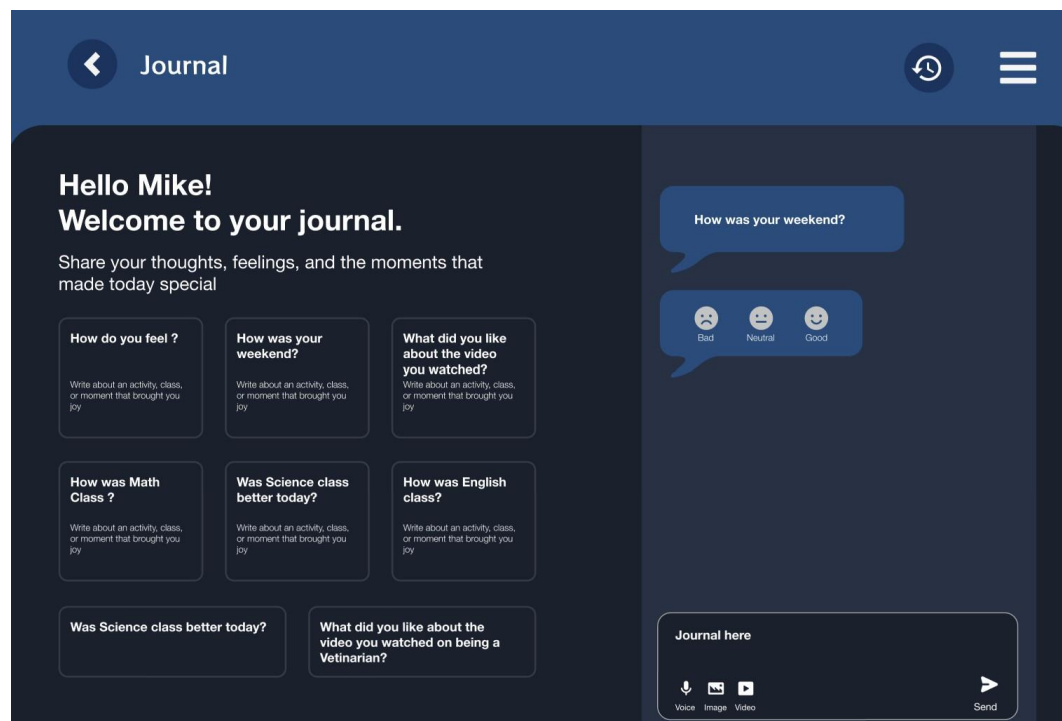
A key feature of this system is its collaborative potential—information about Mike’s selected NorthStar career can be shared with his teacher, enabling educators to incorporate these career interests into personalized learning experiences within the classroom. Furthermore, Mike will be provided with a list of activities to explore, such as visiting a museum or watching relevant YouTube videos, allowing him to gain deeper insights into the career field. This immersive and adaptive system supports both career exploration and personalized learning by giving students tangible, interactive ways to connect with their future goals.



Journaling Section: Integrating AI for Enhanced Engagement

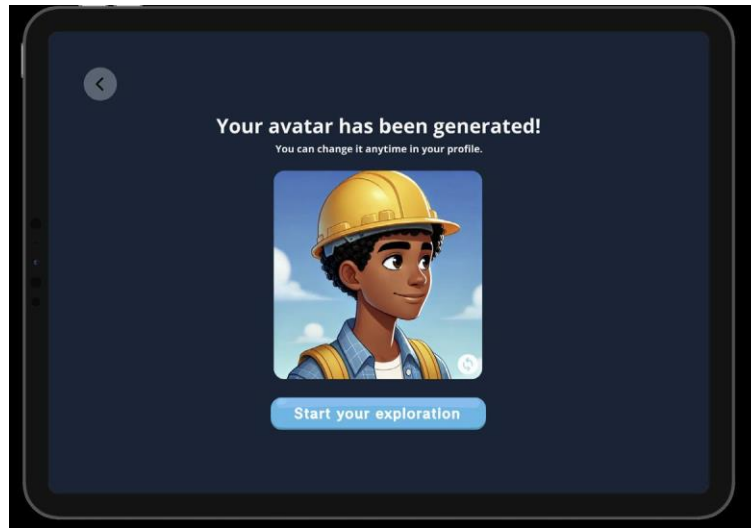
Integrating AI and large language models into the journaling process creates a dynamic and personalized experience, offering students multimodal interaction and real-time engagement while incorporating social and emotional learning, thereby making traditional journaling more engaging and relevant to individual needs. In this design, preset questions are generated based on the student's schedule, such as "How was math class?" or "Did you finish the Civil War section in history?" The research goal is also to seamlessly integrate social and emotional learning (SEL) into these prompts, encouraging students to reflect on both their academic and emotional experiences.

Borrowing from concepts from the learning science community, students have multiple options for responding to these prompts, and they can reply with audio recordings, images, or videos to express how they feel or describe their experiences (Sawyer, 2022). An integrated machine-learning system will analyze these artifacts and engage with the student, making the journaling process more interactive and responsive to individual needs. To further enhance interaction, a chatbot located next to the preset prompts will allow for real-time engagement. The research aims to power this through large language models (LLMs) (though not utilizing OpenAI) in order to curate the data as suggested by (Geburu, 2019). The chatbot can refer to previous conversations to create a more personalized dialogue. This feature not only provides continuity in the journaling experience but also fosters deeper engagement by responding contextually to the student's inputs over time.



Avatar Creation and Identity Representation

The development of the avatar feature will be developed in a future workshop as part of the prototype process. This feature's goal is to allow users to either upload a personal image, take a photo, or describe themselves, creating an avatar that reflects their physical appearance and personal identity. Additionally, users will be prompted to indicate whether they have a specific career aspiration during the onboarding stage. If a career is indicated, the avatar will be designed with a background that represents their chosen occupation, further immersing the student in the career exploration process.



Behind the Prototype: Data-Driven Design and AI Integration

One example of a user interacting with Northstar is a student using the self-reported interest feature. In this case, career recommendations do not rely on academic data (the Scholastic Fit model). For instance, a student who wants to find careers based on their hobbies, like playing video games, would enter their self-reported interest, and the input is matched to job descriptions, skills, and abilities linked to various occupations.

To make this work, OpenAI was used to simplify the descriptions of over 900 occupations, making them kid-friendly and easy for middle school students to understand. This step was required to make career exploration engaging and relatable for young users. However, a limitation of this method is that OpenAI's outputs may carry inherent biases, including those affecting African American males. Over time, as students continue to engage with the prototype, new career descriptions are collected and tailored to represent the interests and experiences of African American boys.

onetsoc_code	title	description	RIASEC...	Activity	teen_description
13-2011.00	Accountants and Auditors	Examine, analyze, and interpret accounting r...	C	Develop...	I like peeping game at money records, gettin...
27-2011.00	Actors	Play parts in stage, television, radio, video, o...	A	Compos...	"I enjoy doing skits on stage or in videos, an...
15-2011.00	Actuaries	Analyze statistical data, such as mortality, ac...	C	Develop...	"I like decoding numbers and facts about thi...
29-1291.00	Acupuncturists	Diagnose, treat, and prevent disorders by sti...	IRS	Build kit...	"I enjoy helping people feel better by using t...
29-1141.01	Acute Care Nurses	Provide advanced nursing care for patients ...	IS	Develop...	"I enjoy learning about how to give top-notc...
25-2059.01	Adapted Physical Education Specialists	Provide individualized physical education ins...	S	Give car...	"I like helping kids and grown-ups who learn...
51-9191.00	Adhesive Bonding Machine Operators and T...	Operate or tend bonding machines that use ...	R	Build kit...	"I like messing around with these cool bondi...
23-1021.00	Administrative Law Judges, Adjudicators, an...	Conduct hearings to recommend or make d...	CE	Develop...	"I like playing the judge in a game where I g...
11-3012.00	Administrative Services Managers	Plan, direct, or coordinate one or more admi...	CE	Develop...	"I enjoy being the boss of stuff around here, ...
25-3011.00	Adult Basic Education, Adult Secondary Educ...	Teach or instruct out-of-school youths and a...	S	Give car...	"I enjoy teaching homies who ain't in school ...
29-1141.02	Advanced Practice Psychiatric Nurses	Assess, diagnose, and treat individuals and f...	IS	Develop...	"I enjoy figuring out what's messing with pe...
11-2011.00	Advertising and Promotions Managers	Plan, direct, or coordinate advertising policie...	E	Manage...	"I enjoy thinking of cool ways to show off pr...
41-3011.00	Advertising Sales Agents	Sell or solicit advertising space, time, or med...	CE	Develop...	"I like pitching ads and getting them placed i...
17-3021.00	Aerospace Engineering and Operations Tech...	Operate, install, adjust, and maintain integra...	CIR	Build kit...	I like messing with computer systems, you k...
17-2011.00	Aerospace Engineers	Perform engineering duties in designing, co...	CIR	Build kit...	"I like designing and building stuff like plane...
13-1011.00	Agents and Business Managers of Artists, Pe...	Represent and promote artists, performers, a...	E	Manage...	"I enjoy being the middleman for cool artists...

After converting the descriptions, I tried a simpler machine learning method called Bag-of-Words (BoW), which counted how often words appeared in job descriptions and stored that information. However, BoW did not consider context, resulting in inconsistent matches. To improve accuracy, I switched to a more advanced method called Bidirectional Encoder Representation from Transformers (BERT). This technique provided better responses and could handle the 900 job descriptions. Below is a code snippet showing how this model processes the inputs and aligns them with specific RIASEC categories based on the student's interests.

```

25
26 # Return the corresponding Title, RAISEC code, and Description
27 results = metadata.iloc[top_indices][['title', 'RIASECCODE', 'description']]
28 return results
29
30 # Example usage
31 if __name__ == "__main__":
32     interest = "I like making food"
33     results = find_closest_matches(interest)
34     print("Top matches based on the interest:")
35     print(results)
36
37
[5] ✓
... Top matches based on the interest:
      title RIASECCODE \
361 Food Scientists and Technologists IR
184 Cooks, Institution and Cafeteria R
360 Food Science Technicians CIR
234 Dietitians and Nutritionists IS
233 Dietetic Technicians S

```

5.2 Next Steps

This phase of my work primarily focused on the prototype design and addressing the challenges related to career-focused technology and its impact on African American males. The next steps will build on this foundation, moving toward prototyping with students who are key contributors to this research. As noted earlier, there are interconnected components involving key stakeholders that need to be incorporated within the prototype.

To support this research, a partnership has been established with Betty Shabazz International School (BSCIS), a charter school in Chicago founded in 1998. BSCIS is part of an African-centered network with two campuses serving grades K-8 in the Chicago area. Earlier this year, BSCIS collaborated with the Design Communication class led by Tomoko Ichikawa, during which teachers were surveyed about the platforms used by the school, and a data dashboard was created as a prototype. My research proposes that information from NorthStar should be integrated into the teacher dashboard to provide insights into students' career aspirations and facilitate personalized learning opportunities based on that information.

IRB#2024-54 has been approved, and consent forms have been provided to BSICS to present to their students. Prototype sessions were scheduled for September but have been pushed back.

The 12-month plan includes:

- Pre-Prototyping: Test the scholastic fit model using simulated data before introducing the prototype to students.
- Workshops: Hold workshops as part of the Co-Design process for the prototype.
- Data Collection: Conduct prototyping sessions with 6th to 8th-grade students at both BSCIS campuses from January to June 2025.
- Data Analysis (Validation): Review qualitative and quantitative data to identify patterns, gain insights, and validate research questions.
- Evaluation & Refinement: Use the six-month data collection window to refine and re-validate findings if necessary.
- Defense

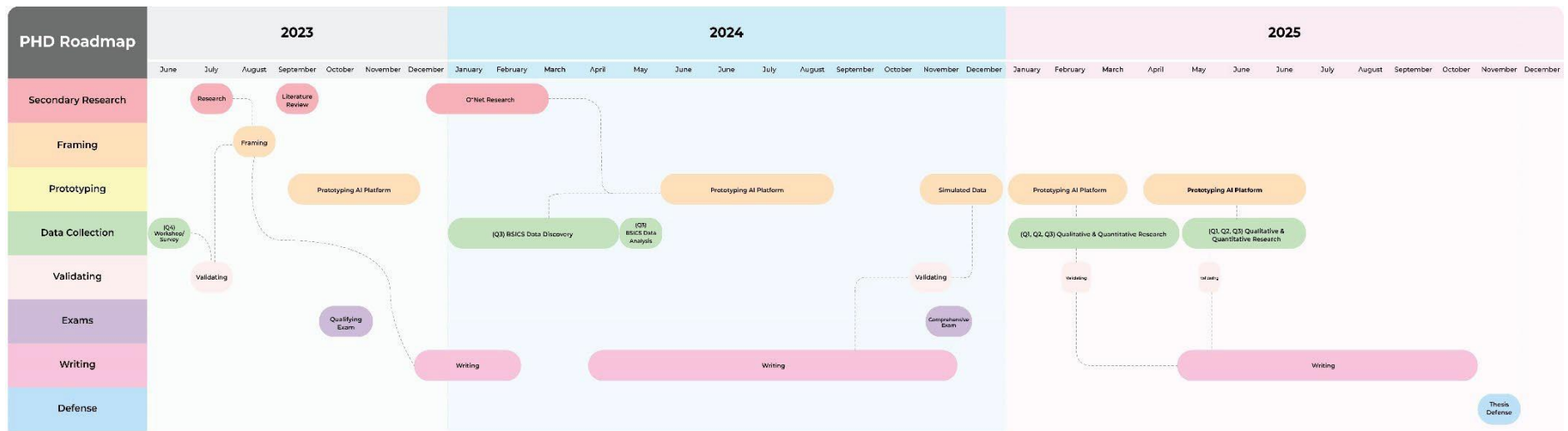


Figure 18- PhD Journey - Next Steps

5.3 Conclusion

By applying Holland's theory, the study creates an abstract foundation for the design of learning technologies that promote career exploration and empowerment, aligning students' career exploration journeys with their personality types and cultural identities. This integration not only helps to make career guidance more personalized but also ensures that marginalized students are provided with tailored opportunities that resonate with their academic, personal, and cultural contexts. These personalized recommendations are continuously refined based on real-time student data, ensuring that students receive up-to-date and relevant career exploration pathways that are aligned with their evolving interests.

Glossary

A:

Adaptive Learning: A computer-based or online educational system that modifies the presentation of material in response to student performance.

Artificial Intelligence (AI) Artifacts: Digital or physical outputs or tools developed using artificial intelligence, serving as manifestations of machine-learning models or systems in practical applications.

Awareness: The conscious knowledge and understanding of various aspects of an environment, situation, or concept in educational settings.

C:

Career-Focused: An approach that emphasizes aligning educational and experiential learning activities with career development and readiness.

Concept Formation: The process through which students learn to categorize information and develop a mental representation of a group based on shared attributes.

Culturally Responsive Design: Design practices that recognize and incorporate cultural backgrounds, perspectives, and experiences into products or services to ensure inclusivity and relevance.

E:

Empowerment: Enabling individuals or groups to have control over factors that affect their lives.

Engagement: The extent to which students are invested in, interested in, and actively involved in learning or activities.

Equity: Ensuring fair treatment, access, opportunity, and advancement for all while striving to identify and eliminate barriers that prevent the full participation of some groups.

Experiential Learning: A process through which students learn by doing and reflecting on their experiences to gain deeper understanding and practical skills.

Exploration Experiences: Opportunities designed for learners to investigate and learn about various topics, fields, or careers through hands-on or interactive engagement.

F:

Formal Learning: Structured, curriculum-based learning provided in educational institutions.

H:

Holland Code: A career assessment model based on John Holland's theory that categorizes individuals and work environments into six personality types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional (RIASEC).

I:

Inclusion: Ensuring individuals, regardless of their background, abilities, or characteristics, are actively and equitably involved and represented in various settings and activities. Environments where everyone feels valued and respected and has equal access to opportunities and resources.

Informal Learning: Unstructured learning that occurs outside of formal education settings, often through life experiences, self-directed exploration, or social interactions.

Interest-based Analysis: The examination of individual preferences and interests to align educational or career recommendations with personal inclinations.

Intersectionality: Considering how various social identities (race, gender, socio-economic status) intersect and impact experiences and outcomes.

L:

Learning Engagement: The degree of attention, curiosity, and interest that students exhibit in their learning activities, contributing to effective learning outcomes.

Learning Technologies: Tools and systems used to facilitate, enhance, or deliver educational content and experiences.

Lived Experience: Building off the foundation of other HCI scholars, lived experience is more than race; it is a social construct that has other cultural groups as a standard norm or oppressive norm. (Egede et al., 2024)

M:

Marginalization: The process through which certain groups are pushed to the edges of society, limiting their access to resources and opportunities.

P:

Personalized Recommendation: A system or process that tailors the experience to individual preferences, interests, hobbies, and academics.

Prompt Engineering: The strategic design and structuring of input prompts to optimize the output of machine learning models, particularly in natural language processing.

Prototype-Based Research: A research approach where prototypes are developed and iteratively tested to study their design, usability, and impact, leading to refinements based on user feedback.

Appendix

Appendix A

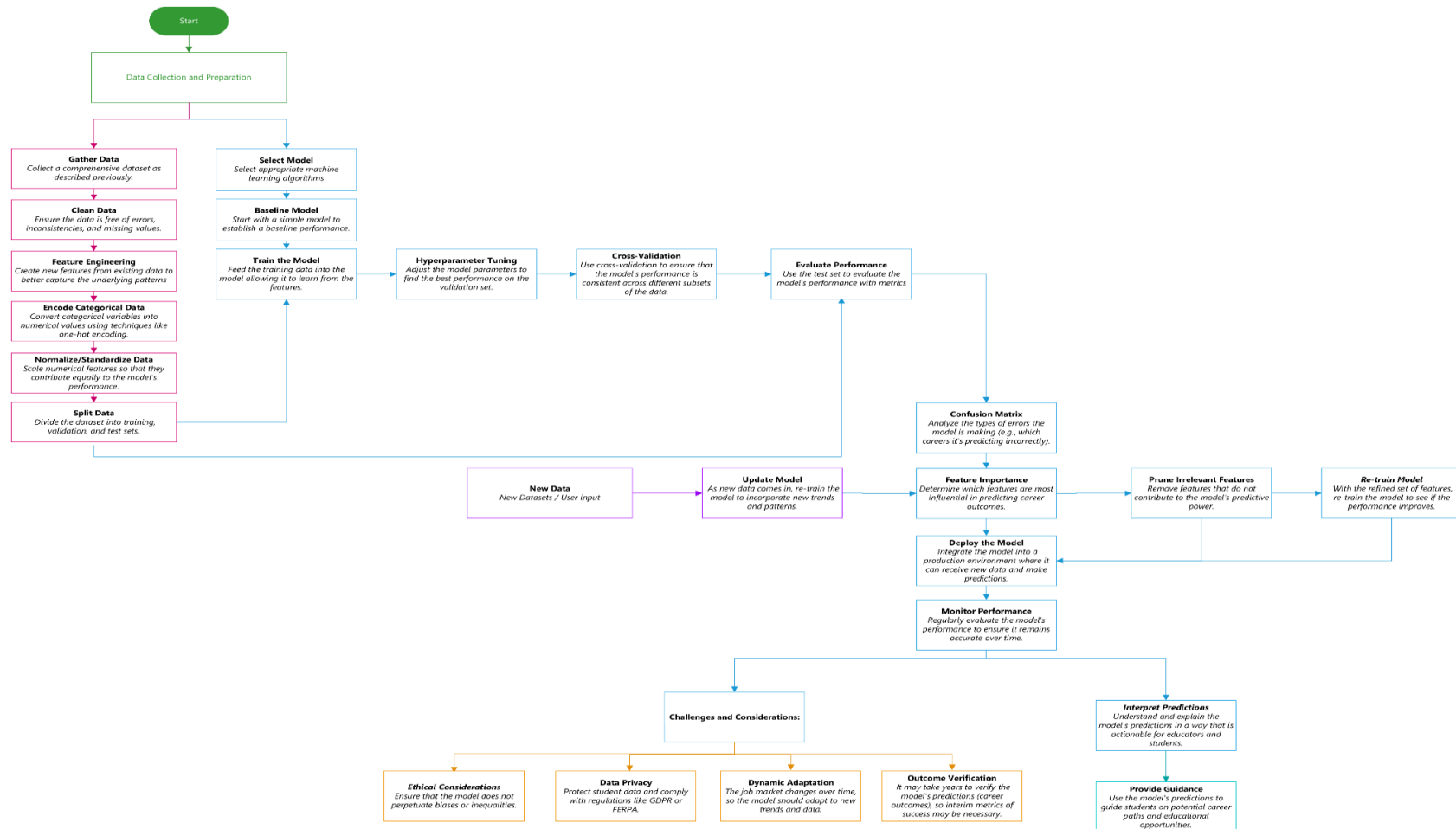


Figure 19 - Learning Pro+ Workflow

Appendix B

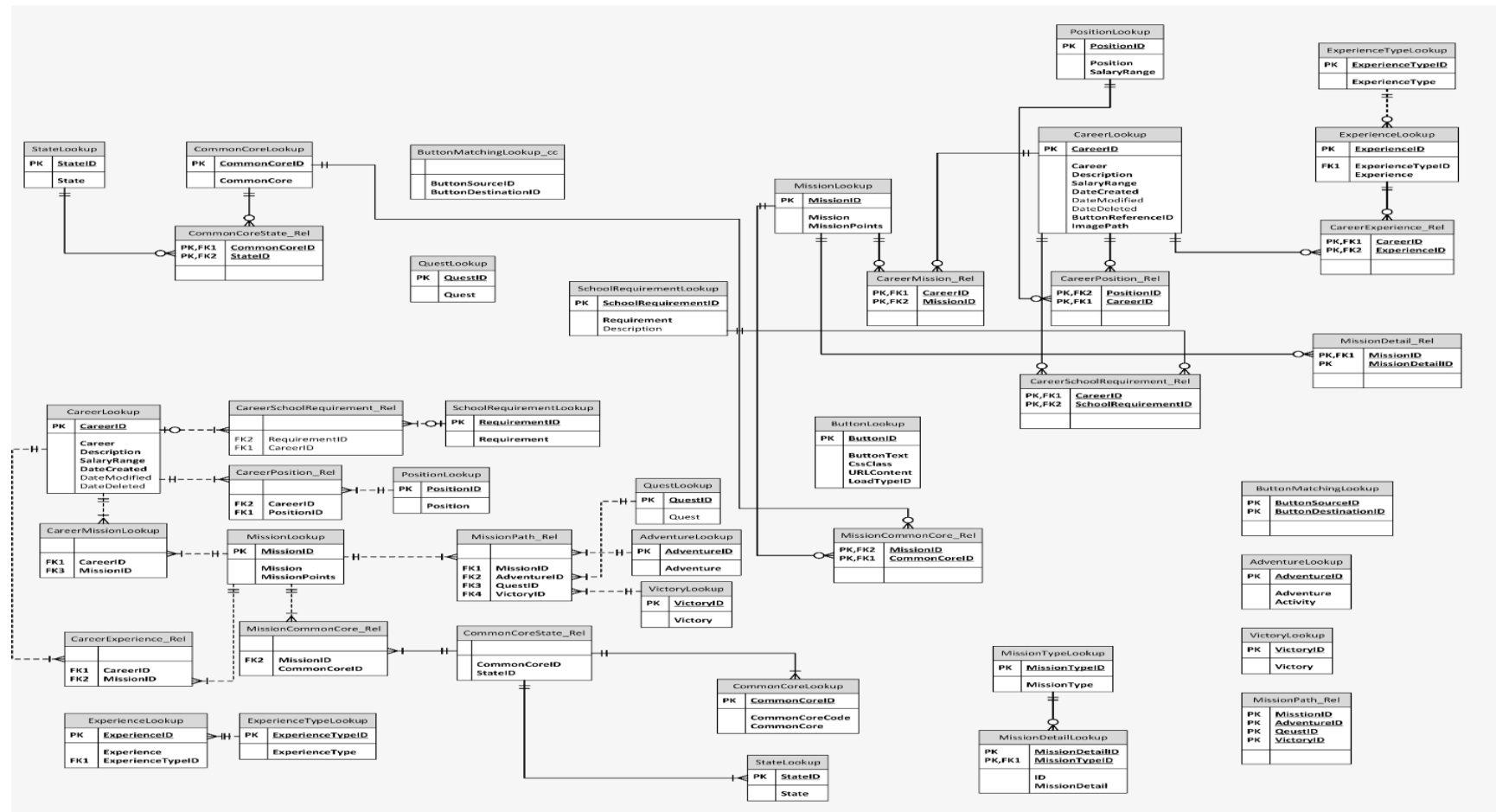


Figure 20 - O*Net Data Model

References

- American School Counselor Association. (2015). *Student-to-school-counselor ratio 2015–2016*.
- Anssi Salonen, Sirpa Kärkkäinen, & Tuula Keinonen. (2018). Career-related instruction promoting students' career awareness and interest towards science learning. *Chemistry Education. Research and Practice*, 19(2), 474–483. <https://doi.org/10.1039/c7rp00221a>
- Arafath, Md. Y., Saifuzzaman, Mohd., Ahmed, S., & Hossain, S. A. (2018). Predicting Career Using Data Mining. *2018 International Conference on Computing, Power and Communication Technologies (GUCON)*, 889–894. <https://doi.org/10.1109/GUCON.2018.8674995>
- Arbona, C., Fan, W., De Dios, M. A., & Olvera, N. (2024). Career decision-making difficulties among career-decided college students. *The Career Development Quarterly*, 72(1), 32–45. <https://doi.org/10.1002/cdq.12339>
- Bacchini, F., & Lorusso, L. (2019). Race, again: How face recognition technology reinforces racial discrimination. *Journal of Information, Communication and Ethics in Society*, 17(3), 321–335. <https://doi.org/10.1108/JICES-05-2018-0050>
- Bang, M., & Vossoughi, S. (2016). Participatory design research and educational justice: Studying learning and relations within social change making. *Cognition and Instruction*, 34(3), 173–193.
- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. <https://doi.org/10.1145/3442188.3445922>
- Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5185–5198. <https://doi.org/10.18653/v1/2020.acl-main.463>
- Benedict, T. (2022). The computer got it wrong: Facial recognition technology and establishing probable cause to arrest. *Wash. & Lee L. Rev.*, 79, 849.
- Benjamin, R. (2019). *Race after technology: Abolitionist tools for the new Jim code*. Polity.
- Brown, C. H., & Knight, D. (2023). Staffing Schools to Support the Classroom: Examining Student-to-School-Counselor Ratios and Academic Student Outcomes in Texas. *Professional School Counseling*, 27(1), 2156759X2311654. <https://doi.org/10.1177/2156759X231165497>
- Bryan, J., Kim, J., & Liu, C. (2022). School counseling college-going culture: Counselors' influence on students' college-going decisions. *Journal of Counseling & Development*, 100(1), 39–55. <https://doi.org/10.1002/jcad.12408>
- Burkett, I. (2012). An introduction to co-design. *Sydney: Knode*, 12.
- Camburn, B., Viswanathan, V., Linsey, J., Anderson, D., Jensen, D., Crawford, R., Otto, K., & Wood, K. (2017). Design prototyping methods: State of the art in strategies, techniques, and guidelines. *Design Science*, 3, e13. <https://doi.org/10.1017/dsj.2017.10>

- Campbell, D. P., & Hansen, J.-I. C. (1981). Manual for the SVIB-SCII: Strong-Campbell interest inventory, form T325 of the Strong vocational interest blank. (*No Title*).
- Caratozzolo, P., Cukierman, U., Nørgaard, B., Schrey-Niemenmaa, K., Azofeifa, J. D., & Rueda-Castro, V. (2024). Future Skills Forecasting: Ensuring Quality Learning for Every Segment of the Workforce. *2024 IEEE Global Engineering Education Conference (EDUCON)*, 1–5. <https://doi.org/10.1109/EDUCON60312.2024.10578916>
- Chen, S., Chen, H., Ling, H., & Gu, X. (2021). How Do Students Become Good Workers? Investigating the Impact of Gender and School on the Relationship between Career Decision-Making Self-Efficacy and Career Exploration. *Sustainability*, *13*(14), 7876. <https://doi.org/10.3390/su13147876>
- Chen, Z., & Schmidt, R. (2024). *Exploring a Behavioral Model of “Positive Friction” in Human-AI Interaction* (arXiv:2402.09683). arXiv. <http://arxiv.org/abs/2402.09683>
- Cheng, A. Y., Guo, M., Ran, M., Ranasaria, A., Sharma, A., Xie, A., Le, K. N., Vinaithirthan, B., Luan, S., & Wright, D. T. H. (2024). *Scientific and Fantastical: Creating Immersive, Culturally Relevant Learning Experiences with Augmented Reality and Large Language Models*. 1–23.
- Cheng, M.-M., Lacaste, A. V., Saranza, C., & Chuang, H.-H. (2021). Culturally Responsive Teaching in Technology-Supported Learning Environments in Marine Education for Sustainable Development. *Sustainability*, *13*(24), 13922. <https://doi.org/10.3390/su132413922>
- Chiesa, R., Massei, F., & Guglielmi, D. (2016). Career Decision-Making Self-Efficacy Change in Italian High School Students. *Journal of Counseling & Development*, *94*(2), 210–224. <https://doi.org/10.1002/jcad.12077>
- Comendador, B. E. V., Rabago, L. W., & Tanguilig, B. T. (2016). An educational model based on Knowledge Discovery in Databases (KDD) to predict learner’s behavior using classification techniques. *2016 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC)*, 1–6. <https://doi.org/10.1109/ICSPCC.2016.7753623>
- Compton, J. (2021). *African American girls: A critical approach to preparation for digital arts careers* [Northeastern University]. <https://doi.org/10.17760/D20413978>
- Conley, D. T. (2012). A Complete Definition of College and Career Readiness. *Educational Policy Improvement Center (NJI)*.
- Contrino, M. F., Reyes-Millán, M., Vázquez-Villegas, P., & Membrillo-Hernández, J. (2024). Using an adaptive learning tool to improve student performance and satisfaction in online and face-to-face education for a more personalized approach. *Smart Learning Environments*, *11*(1), 6. <https://doi.org/10.1186/s40561-024-00292-y>
- Converse, P. D., Oswald, F. L., Gillespie, M. A., Field, K. A., & Bizot, E. B. (2004). MATCHING INDIVIDUALS TO OCCUPATIONS USING ABILITIES AND THE O*NET: ISSUES AND AN APPLICATION IN CAREER GUIDANCE. *Personnel Psychology*, *57*(2), 451–487. <https://doi.org/10.1111/j.1744-6570.2004.tb02497.x>
- Costanza-Chock, S. (2020). *Design Justice: Community-Led Practices to Build the Worlds We Need*. The MIT Press. <https://doi.org/10.7551/mitpress/12255.001.0001>

- Cristol, D., Choi, M., Mitchell, R., & Burbidge, J. (2015). Mobile Technology in K-12 Environments. In Y. Zhang (Ed.), *Handbook of Mobile Teaching and Learning* (pp. 669–682). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-54146-9_33
- Cureton, A., & Aguinaldo, E. (2024). After High School, What’s Next? Exploring Refugee Youths’ Perceptions and Preparations Around College Choice and Transition. *Journal of Career Assessment*, 32(1), 182–201. <https://doi.org/10.1177/10690727231185175>
- Curry, J. R., Belser, C. T., & Binns, I. C. (2013). Integrating Postsecondary College and Career Options in the Middle Level Curriculum: Middle Level Educators Must Find Ways to Integrate Postsecondary Education and Career Exploration Throughout the Curriculum. *Middle School Journal*, 44(3), 26–32. <https://doi.org/10.1080/00940771.2013.11461852>
- Dancy, C. L., & Saucier, P. K. (2022). AI and Blackness: Toward Moving Beyond Bias and Representation. *IEEE Transactions on Technology and Society*, 3(1), 31–40. <https://doi.org/10.1109/TTS.2021.3125998>
- Darley, J. G., & Harenah, T. (1955). *Vocational interest measurement: Theory and practice*.
- Day, S. X., & Rounds, J. (1998). Universality of vocational interest structure among racial and ethnic minorities. *American Psychologist*, 53(7), 728.
- De Caro-Barek, V., Lysne, D. A., Støckert, R., Solbjørg, O. K., & Flem Røren, K. A. (2024). Places of Learning: A Discussion of Radcliffe’s Framework for Learning Spaces in Light of Merleau-Ponty’s Phenomenology. *Nordic Studies in Education*, 44(1). <https://doi.org/10.23865/nse.v44.5654>
- Demartini, C. G., Sciascia, L., Bosso, A., & Manuri, F. (2024). Artificial Intelligence Bringing Improvements to Adaptive Learning in Education: A Case Study. *Sustainability*, 16(3), 1347. <https://doi.org/10.3390/su16031347>
- Dinampo, R. F., Fatura, F. E., & Paleracio, J. B. (2024). Correlating Students’ personality Types with Curriculum Exits in Paliparan Iii Senior High School. *Ignatian International Journal for Multidisciplinary Research*, 2(2), 265–285.
- Dozier, V. C., Morgan, M., Burbrink, I., & Peace, C. (2022). Standardized career course curriculum: Effects on negative career thoughts. *The Career Development Quarterly*, 70(4), 314–326. <https://doi.org/10.1002/cdq.12305>
- Drame, E. R., & Irby, D. J. (Eds.). (2016). *Black participatory research: Power, identity, and the struggle for justice in education*. Palgrave Macmillan.
- Dutta, I., Shah, A., Safa, Ms. M., & Jayavel, K. (2022). Building a Recommendation System for Career Advice for Students from Professionals. *2022 International Conference on Computer Communication and Informatics (ICCCI)*, 1–10. <https://doi.org/10.1109/ICCCI54379.2022.9740837>
- Ege, D. N., Goudswaard, M., Gopsill, J., Steinert, M., & Hicks, B. (2024). What, how and when should I prototype? An empirical study of design team prototyping practices at the IDEA challenge hackathon. *Design Science*, 10, e22. <https://doi.org/10.1017/dsj.2024.16>
- Egede, L., Coney, L., Johnson, B., Harrington, C., & Ford, D. (2024). “For Us By Us”: Intentionally Designing Technology for Lived Black Experiences. *Designing Interactive Systems Conference*, 3210–3224. <https://doi.org/10.1145/3643834.3661535>

Ersoy, N. S. (2023). Empowering Inclusion: Addressing Barriers in Distance Learning for Disadvantaged Groups. *Turkish Online Journal of Educational Technology-TOJET*, 22(4), 52–60.

Faculty of Teacher Training and Education, University of Lampung, Indonesia, & Dahlan, S. (2024). Vocational Personality Profile Based on Where Students Live and School Popularity: Classification of the Holland Model in Lampung Society, Indonesia. *INTERNATIONAL JOURNAL OF MULTIDISCIPLINARY RESEARCH AND ANALYSIS*, 07(05). <https://doi.org/10.47191/ijmra/v7-i05-05>

Fernández-Nistal, M. T., Mora-Soto, J. K., & Ponce-Zaragoza, F. A. (2024). Congruence between interests and occupational aspirations: Its relationship with personality and self-efficacy in a sample of high school students in Mexico. *Electronic Journal of Research in Educational Psychology*, 22(62).

Firkola, P. (2021). Review of Career Assessment Tools. *Archives of Business Research*, 9(11), 131–140. <https://doi.org/10.14738/abr.911.11299>

Fössing, E., & Drechsler, J. (2024). *An Evaluation of Synthetic Data Generators Implemented in the Python Library Synthcity*. 178–193.

Foster, C. (2024). Methodological pragmatism in educational research: From qualitative-quantitative to exploratory-confirmatory distinctions. *International Journal of Research & Method in Education*, 47(1), 4–19. <https://doi.org/10.1080/1743727X.2023.2210063>

Freeman, J. A., & Hirsch, B. T. (2008). College majors and the knowledge content of jobs. *Economics of Education Review*, 27(5), 517–535.

Gebru, T. (2019). *Oxford Handbook on AI Ethics Book Chapter on Race and Gender*. <https://doi.org/10.48550/ARXIV.1908.06165>

Gottfredson, G. (1996). Dictionary of Holland occupational codes. *Psychological Assessment Resources*.

Gregory, C., & Lewis, P. (2016). Linking client assessment profiles to O* NET® occupational profiles within the O* NET interest profiler short form and mini interest profiler (Mini-IP). *National Center for O* NET Development*. https://www.onetcenter.org/Reports/Mini-IP_Linking.Html.

Han, S., Kelly, E., Nikou, S., & Svec, E.-O. (2022). Aligning artificial intelligence with human values: Reflections from a phenomenological perspective. *AI & SOCIETY*, 37(4), 1383–1395. <https://doi.org/10.1007/s00146-021-01247-4>

Hansen, J. (1994). The measurement of vocational interests. *Personnel Selection and Classification*, 293–316.

Harrington, C., Erete, S., & Piper, A. M. (2019). Deconstructing Community-Based Collaborative Design: Towards More Equitable Participatory Design Engagements. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW), 1–25. <https://doi.org/10.1145/3359318>

Hilts, D., Liu, Y., & Guo, X. (2023). Student-To-School-Counselor Ratios, School-Level Factors, and Leadership Practices of School Counselors: A National Investigation. *Professional School Counseling*, 27(1), 2156759X2311821. <https://doi.org/10.1177/2156759X231182135>

Hogan, R., & Blake, R. (1999). John Holland's Vocational Typology and Personality Theory. *Journal of Vocational Behavior*, 55(1), 41–56. <https://doi.org/10.1006/jvbe.1999.1696>

- Holland, J. (1997). Making vocational choices: A theory of vocational personalities and work environments. *Psychological Assessment Resources*.
- Holland, J. L. (1966). *The psychology of vocational choice: A theory of personality types and model environments*.
- Holland, J. L. (1985). *Vocational preference inventory*. Consulting Psychologists Press.
- Holland, J. L., & Gottfredson, G. D. (1976). Using a typology of persons and environments to explain careers: Some extensions and clarifications. *The Counseling Psychologist*, 6(3), 20–29.
- Holland, J. L., Powell, A. B., & Fritzsche, B. A. (1997). *The self-directed search (SDS): Professional user's guide*. Psychological Assessment Resources.
- Huang, X., Sang, J., & Xu, C. (2022). Image-Based personality questionnaire design. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, 18(4), 1–20.
- Hussain, A., Hussain, S., & Ali, Z. (2021). THE IMPACT OF CONGRUENCE ON STUDENT ACADEMIC PERFORMANCE: ASSESSMENT OF HOLLAND THEORY IN NON-WESTERN CULTURE. *Humanities & Social Sciences Reviews*, 9(3), 81–91. <https://doi.org/10.18510/hssr.2021.939>
- I George, R., & Paul, B. (2024). Unravelling the interplay between competencies, career preparedness, and perceived employability among postgraduate students: A structural model analysis. *Asia Pacific Education Review*, 25(2), 439–457. <https://doi.org/10.1007/s12564-023-09896-4>
- Jackson, D. N., Sehgal, N., & Baur, C. (2022). Benefits of mHealth Co-design for African American and Hispanic Adults: Multi-Method Participatory Research for a Health Information App. *JMIR Formative Research*, 6(3), e26764. <https://doi.org/10.2196/26764>
- Jadallah, C., Ballard, H., Meyer, R. M., Ching, C. C., & Patterson Williams, A. (2024). “Seeing power” between young people and conservation professionals in the design of a community-based watershed monitoring initiative. *Journal of the Learning Sciences*, 33(1), 1–40. <https://doi.org/10.1080/10508406.2024.2310240>
- Janise S. Parker, Jasmine N. Garnes, Emily D. Oliver, Avery Amabile, & Sarathy, A. (2020). It Takes a Village: Understanding African American High School Students’ Self-Determination in School. *School Psychology Review*, 49(2), 111–129. <https://doi.org/10.1080/2372966X.2020.1717371>
- Jelonek, M., & Urbaniec, M. (2019). Development of Sustainability Competencies for the Labour Market: An Exploratory Qualitative Study. *Sustainability*, 11(20). <https://doi.org/10.3390/su11205716>
- Johnson, K. F., Robins, L. B., Derlan Williams, C., & Townsend, T. G. (2021). Interrogating Systems That Cause Disparities: Testing the Social-Ecological Model in Low- Versus High-Density African American Communities. *Journal of Social, Behavioral, and Health Sciences*, 15(1). <https://doi.org/10.5590/JSBHS.2021.15.1.22>
- Jones, M. (2020). *EFFECTS OF AN ANCHORED CAREER-FOCUSED CURRICULUM ON THE KNOWLEDGE AND ATTITUDES OF AT-RISK HIGH SCHOOL STUDENTS* [[object Object]]. <https://doi.org/10.13023/ETD.2020.122>
- Khan, B., Jan, S., Khan, W., & Chughtai, M. I. (2024). An Overview of ETL Techniques, Tools, Processes and Evaluations in Data Warehousing. *Journal on Big Data*, 6(1), 1–20. <https://doi.org/10.32604/jbd.2023.046223>

- Koçak, O., Ak, N., Erdem, S. S., Sinan, M., Younis, M. Z., & Erdoğan, A. (2021). The Role of Family Influence and Academic Satisfaction on Career Decision-Making Self-Efficacy and Happiness. *International Journal of Environmental Research and Public Health*, 18(11), 5919. <https://doi.org/10.3390/ijerph18115919>
- Kolb, A. Y., & Kolb, D. A. (2009). Experiential learning theory: A dynamic, holistic approach to management learning, education and development. *The SAGE Handbook of Management Learning, Education and Development*, 7(2), 42–68.
- Koys, D. J. (2017). Using the Department of Labor’s “My Next Move” to Improve Career Preparedness. *Journal of Management Education*, 41(1), 94–117. <https://doi.org/10.1177/1052562916670195>
- Lateef, H., Gale, A., Jellesma, F., & Borgstrom, E. (2024). “The Belief to Aspire”: The Association Between Afrocentric Values in the Educational and Career Aspirations of Young Black Males. *The Urban Review*, 56(1), 19–34.
- Lau, P. L., Anctil, T., Ee, G. T., Jaafar, J. L. S., & Kin, T. G. (2020). Self-Concept, Attitudes Toward Career Counseling, and Work Readiness of Malaysian Vocational Students. *The Career Development Quarterly*, 68(1), 18–31. <https://doi.org/10.1002/cdq.12210>
- Layton, W. L. (1958). *Counseling use of the Strong Vocational Interest Blank*.
- Lehtinen, A., & Raerinne, J. (2023). Simulated Data in Empirical Science. *Foundations of Science*. <https://doi.org/10.1007/s10699-023-09934-9>
- Malin, J. R., Bragg, D. D., & Hackmann, D. G. (2017). College and Career Readiness and the Every Student Succeeds Act. *Educational Administration Quarterly*, 53(5), 809–838. <https://doi.org/10.1177/0013161X17714845>
- Marinai, S. (2008). Introduction to Document Analysis and Recognition. In S. Marinai & H. Fujisawa (Eds.), *Machine Learning in Document Analysis and Recognition* (Vol. 90, pp. 1–20). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-76280-5_1
- Mazzotti, V. L., Morningstar, M. E., Lombardi, A., Kwiatek, S., Taconet, A., Buddeke, K., Monahan, J., & Harris, R. (2024). Policy and Practice Considerations to Support College and Career Readiness for Youth With Disabilities: A Systematic Mixed Studies Review. *Journal of Disability Policy Studies*, 34(4), 278–289. <https://doi.org/10.1177/10442073221130528>
- McFarlane-Beau, J. G. (2021). *Phenomenology of Social Justice on Leadership Development of*.
- Mejeh, M., & Rehm, M. (2024). Taking adaptive learning in educational settings to the next level: Leveraging natural language processing for improved personalization. *Educational Technology Research and Development*, 72(3), 1597–1621. <https://doi.org/10.1007/s11423-024-10345-1>
- Mhide, K. M., Obinne, A. D. E., Aduloju, M. O., & Agi, C. I. (PhD). (2023). Effect of No-Zero Grading Policy on Achievement, Self-Efficacy and Attitude towards Physics in Secondary Schools in Benue North-East Senatorial District, Benue State. *International Journal of Research and Innovation in Social Science*, VII(X), 1306–1318. <https://doi.org/10.47772/IJRISS.2023.701101>
- Michael Healy, Sara Hammer, & Peter McIlveen. (2020). Mapping graduate employability and career development in higher education research: A citation network analysis. *Studies in Higher Education*, 47(4), 799–811. <https://doi.org/10.1080/03075079.2020.1804851>

- Mishra, A., Sahay, A., Pandey, M. anjusha, & Routaray, S. S. (2023). News text Analysis using Text Summarization and Sentiment Analysis based on NLP. *2023 3rd International Conference on Smart Data Intelligence (ICSMDI)*, 28–31. <https://doi.org/10.1109/ICSMDI57622.2023.00014>
- Mittal, V., Gangodkar, D., & Pant, B. (2020). Exploring The Dimension of DNN Techniques For Text Categorization Using NLP. *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*, 497–501. <https://doi.org/10.1109/ICACCS48705.2020.9074228>
- Naeem, M., Ozuem, W., Howell, K., & Ranfagni, S. (2023). A Step-by-Step Process of Thematic Analysis to Develop a Conceptual Model in Qualitative Research. *International Journal of Qualitative Methods*, 22, 16094069231205789. <https://doi.org/10.1177/16094069231205789>
- Neelamani, samal, Nandini, B., Abhishek, C., Rahul, D., Yuvjeet, & Mohali, P. (2023). USING MACHINE LEARNING FOR CAMPUS PLACEMENT PREDICTION. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, 07(11), 1–11. <https://doi.org/10.55041/IJSREM27027>
- Noble, S. U. (2020). *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York University Press. <https://doi.org/10.18574/nyu/9781479833641.001.0001>
- Nye, C. D., Prasad, J., & Rounds, J. (2021). The effects of vocational interests on motivation, satisfaction, and academic performance: Test of a mediated model. *Journal of Vocational Behavior*, 127, 103583. <https://doi.org/10.1016/j.jvb.2021.103583>
- Ogbonnaya-Ogburu, I. F., Smith, A. D., To, A., & Toyama, K. (2020). *Critical race theory for HCI*. 1–16.
- Paay, J., Taffe, S., & Pedell, S. (2023). Building relationships with remote participants through playful technology interactions in online codesign. *Design and Technology Education: An International Journal*, 28(1), 77–99.
- Panditharathna, P. M. R. A., & Rajapaksha, S. (2023). Question and Answering System For Investment Promotion Based on NLP. *2023 International Conference for Advancement in Technology (ICONAT)*, 1–7. <https://doi.org/10.1109/ICONAT57137.2023.10080255>
- Pardi, P. (2024). Utilization of Riasec Theory to Identify Career Personalities in Deaf Students: A Systematic Review. *Mutiara: Multidisciplinary Scientific Journal*, 2(5), 296–305. <https://doi.org/10.57185/mutiara.v2i5.195>
- Peschard, I. (2011). *Is Simulation an Epistemic Substitute for Experimentation?*
- Peurach, D. J., Cohen, D. K., Yurkofsky, M. M., & Spillane, J. P. (2019). From Mass Schooling to Education Systems: Changing Patterns in the Organization and Management of Instruction. *Review of Research in Education*, 43(1), 32–67. <https://doi.org/10.3102/0091732X18821131>
- Potvin, A. S., Teeters, L. P., & Penuel, W. R. (2021). *Designing for compassion in schools: A humanizing approach to co-design*. Proceedings of the 15th International Conference of the Learning Sciences-ICLS 2021.
- Potvin, A. S., Teeters, L. P., Penuel, W. R., & Dimidjian, S. (2024). Humanizing Co-design through attention to educators' affective and relational experiences. *Journal of the Learning Sciences*, 33(1), 41–79. <https://doi.org/10.1080/10508406.2024.2318557>

- Putka, D. J., Dahlke, J. A., Burke, M. I., Rounds, J., & Lewis, P. (n.d.). *Using Machine Learning to Develop Occupational Interest Profiles and High-Point Codes for the O*NET System*.
- R. Honra, J., C. Monterola, S. L., & T. Yangco, R. (2024). Career-focused teaching and its effects on students' biology-technical-vocational-fused skills. *International Journal of Evaluation and Research in Education (IJERE)*, 13(5), 3427. <https://doi.org/10.11591/ijere.v13i5.29131>
- Raines, T. C., & Talapatra, D. (2019). College and Career Readiness Consultation for High-Risk Youth: An Introduction. *Journal of Educational and Psychological Consultation*, 29(3), 255–259. <https://doi.org/10.1080/10474412.2019.1565540>
- Reed, M., Castillo, I., Georgia, B. W., Glass, H. L., Ryan, D., & Cygan, H. R. (2024). An Integrative Review of College Readiness Programs for Black High School Students; Opportunities for School Nurse Involvement. *The Journal of School Nursing*, 40(1), 26–42. <https://doi.org/10.1177/10598405221105012>
- Robinson, N. K., Meyer, D., Prince, J. P., McLean, C., & Low, R. (2000). Mining the internet for career information: A model approach for college students. *Journal of Career Assessment*, 8(1), 37–54.
- Rosen, D., Holmberg, K., & Holland, J. L. (1994). *The educational opportunities finder: For use with the Self-Directed Search and the Vocational Preference Inventory*. Psychological Assessment Resources.
- Rounds, J., Phan, W. M. J., Amrhein, R., & Lewis, P. (n.d.). *Examining the Efficacy of Emoji Anchors for the O*NET Interest Profiler Short Form*.
- Rounds, J., Smith, T., Hubert, L., Lewis, P., & Rivkin, D. (1999). Development of occupational interest profiles for O* NET. *Raleigh, NC: National Center for O* NET Development*.
- Salako, A. A., & Adediran, O. A. (2024). Choice of Career Gap Among Adolescents with Disabilities. *European Journal of Training and Development Studies*, 11(1), 40–51.
- Sawyer, R. K. (Ed.). (2022). *The Cambridge handbook of the learning sciences* (Third Edition). Cambridge University Press.
- Shelehov, I. V., Prylepa, D. V., Khibovska, Yu. O., & Otroshchenko, M. S. (2023). MACHINE LEARNING DECISION SUPPORT SYSTEMS FOR ADAPTATION OF EDUCATIONAL CONTENT TO THE LABOR MARKET REQUIREMENTS. *Radio Electronics, Computer Science, Control*, 1, 62. <https://doi.org/10.15588/1607-3274-2023-1-6>
- Singha, R., & Singha, S. (2023). Educational Innovation Transforming Higher Education for Workforce Readiness: In B. Christiansen & A. M. Even (Eds.), *Advances in Higher Education and Professional Development* (pp. 37–55). IGI Global. <https://doi.org/10.4018/979-8-3693-0517-1.ch003>
- Slaten, C. D., Wadley, K., Harris, P. C., Sebastian, B., Lee, J., & Curs, B. R. (2024). “We Try to Build Relationships”: The Role of School Community and Belonging on Career Preparedness. *Journal of Career Development*, 08948453241271375. <https://doi.org/10.1177/08948453241271375>
- Strobl, C., & Leisch, F. (2024). Against the “one method fits all data sets” philosophy for comparison studies in methodological research. *Biometrical Journal*, 66(1), 2200104. <https://doi.org/10.1002/bimj.202200104>
- Su, R. (2020). The three faces of interests: An integrative review of interest research in vocational, organizational, and educational psychology. *Journal of Vocational Behavior*, 116, 103240. <https://doi.org/10.1016/j.jvb.2018.10.016>

SUPER, D. (1985). *MAKING VOCATIONAL CHOICES-A THEORY OF VOCATIONAL PERSONALITIES AND WORK ENVIRONMENTS*, -HOLLAND, J.L.

Taddeo, M., Blanchard, A., & Thomas, C. (2024). From AI Ethics Principles to Practices: A Teleological Methodology to Apply AI Ethics Principles in The Defence Domain. *Philosophy & Technology*, 37(1), 42. <https://doi.org/10.1007/s13347-024-00710-6>

Tan, Z., Huang, S., Jia, Z., Cai, J., Li, Y., Lu, W., Zhuang, Y., Tu, K., Xie, P., Huang, F., & Jiang, Y. (2023). *DAMO-NLP at SemEval-2023 Task 2: A Unified Retrieval-augmented System for Multilingual Named Entity Recognition* (arXiv:2305.03688). arXiv. <http://arxiv.org/abs/2305.03688>

Taqyah, C., Akhnaf, A. F., & Dumbi, K. S. (2024). Structural Validity of Eureka Interest Inventory among Indonesian High School Student: An Analysis on Holland's RIASEC Model. *Psikologika: Jurnal Pemikiran Dan Penelitian Psikologi*, 29(1), 1–18.

Thiem, K. C., & Dasgupta, N. (2022). From Precollege to Career: Barriers Facing Historically Marginalized Students and Evidence-Based Solutions. *Social Issues and Policy Review*, 16(1), 212–251. <https://doi.org/10.1111/sipr.12085>

Thompson, K. R., & Koys, D. J. (2010). The management curriculum and assessment journey: Use of Baldrige criteria and the occupational network database. *Journal of Leadership & Organizational Studies*, 17(2), 156–166.

Thompson, N., Erete, S., Nacu, D., & Pinkard, N. (2023). Designing toward justice: Making space for Black and Latine parent involvement in youth OST computing and STEAM education. *International Journal of Child-Computer Interaction*, 35, 100548. <https://doi.org/10.1016/j.ijcci.2022.100548>

Tracey, T. J., & Rounds, J. B. (1993). Evaluating Holland's and Gati's vocational-interest models: A structural meta-analysis. *Psychological Bulletin*, 113(2), 229.

Vaughn, O. (2023). *Three Sides to Success: Exploring How Inclusive Partnerships Can Nurture Robust Family and Community Engagement in the Classroom for African-American Students* [Master of Science, Dominican University of California]. <https://doi.org/10.33015/dominican.edu/2023.EDU.14>

Walonoski, J., Kramer, M., Nichols, J., Quina, A., Moesel, C., Hall, D., Duffett, C., Dube, K., Gallagher, T., & McLachlan, S. (2018). Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association*, 25(3), 230–238. <https://doi.org/10.1093/jamia/ocx079>

Wang, L., Lyu, C., Ji, T., Zhang, Z., Yu, D., Shi, S., & Tu, Z. (2023). *Document-Level Machine Translation with Large Language Models* (arXiv:2304.02210). arXiv. <http://arxiv.org/abs/2304.02210>

Williams, D. P. (2021). Constructing Situated and Social Knowledge: Ethical, Sociological, and Phenomenological Factors in Technological Design. In Z. Pirtle, D. Tomblin, & G. Madhavan (Eds.), *Engineering and Philosophy* (Vol. 37, pp. 143–159). Springer International Publishing. https://doi.org/10.1007/978-3-030-70099-7_7

Williams, S. E., Fergus, T. A., & Ginty, A. T. (2024). Development and Validation of the Ease of Imagery Questionnaire. *Assessment*, 10731911241260233. <https://doi.org/10.1177/10731911241260233>

Yao Cheng & Yu Liang. (2023). The Development of Artificial Intelligence in Career Initiation Education and Implications for China. *European Journal of Artificial Intelligence and Machine Learning*, 2(4), 4–10. <https://doi.org/10.24018/ejai.2023.2.4.32>

Zahra, S. T., & Malik, A. A. (2024). High School Students' Career Program and Parental Involvement: Effect on Career Maturity. *Journal of Behavioural Sciences*, 34(1).

Zahrebniuk, Y. (2023). The System of Career Guidance for High School Students in the United States of America. *Educational Challenges*, 28(1), 188–198. <https://doi.org/10.34142/2709-7986.2023.28.1.15>

Zainul, M., & Maskur, M. (2024). MANAGING HIGHER EDUCATION: ENHANCING STUDENT LOYALTY IN A CAREER-FOCUSED ERA. *Polish Journal of Management Studies*, 29(1), 372–387. <https://doi.org/10.17512/pjms.2024.29.1.22>

Zicari, R. V., Brusseau, J., Blomberg, S. N., Christensen, H. C., Coffee, M., Ganapini, M. B., Gerke, S., Gilbert, T. K., Hickman, E., Hildt, E., Holm, S., Kühne, U., Madai, V. I., Osika, W., Spezzatti, A., Schnebel, E., Tithi, J. J., Vetter, D., Westerlund, M., ... Kararigas, G. (2021). On Assessing Trustworthy AI in Healthcare. Machine Learning as a Supportive Tool to Recognize Cardiac Arrest in Emergency Calls. *Frontiers in Human Dynamics*, 3, 673104. <https://doi.org/10.3389/fhumd.2021.673104>